Polished theorems: statements in coordinates, proofs, neighbourhoods, cells, status

Prepared for Ignacio López and Gabriela Araujo-Pardo. Draft of 2026-10-25 (job thm-polish, branch `claude/thm-polish-2oddll`). Sources: `research/NOTE.md`, `research/geometry/geometry.tex` and `PROOFREAD.md`, `research/plus2/coords/coords.tex`, `research/p4/regular/REPORT.md`, `research/theorems/REGULARITIE` `research/gq4*`, `lean/`. Nothing here is a new result: this file restates the lab's earlier theorems with complete proofs, writes every construction in projective coordinates, and lists the neighbourhood of every new vertex so that degrees and diameter can be checked by hand.

For each result we give: **Statement** (in coordinates), **New vertices** (exact neighbourhoods), **Proof**, **Cells** (the 2013 table value against ours), **Status**. Status words:

- *paper proof*: a complete written proof (the one given here);
- *computer-checked*: the graphs were built from the definition and checked (bipartite, degrees, exact diameter by breadth-first search from every vertex), by at least two independent programs;
- *Lean-checked*: proved in Lean 4 with Mathlib (Lean 4.34.1). `#print axioms` lists only `propext`, `Classical.choice`, `Quot.sound`; there is no `sorry`, `admit`, new `axiom` or `native_decide` in the library. Section 9 says exactly which statements are covered.

Changes from the earlier drafts

Errors and corrections found while preparing this version (none affects a theorem or a record value):

1. **The 2013 baseline at $D = 3$.** The README and the values table of `geometry.tex` list $(13, 3) = 274$ and $(15, 3) = 374$ as "table" values. Those are the later values of Araujo-Pardo and López [AL20] (Graphs Combin. 2022). The 2013 survey table (Miller–Širáň, Electron. J. Combin. DS14v2, Table 4) has 270 and 370 there. Both columns are now shown, here and in `geometry.tex` (Table 4). Provenance of the current values: the wiki was read on 2026-10-10 for $(7, 4)$, $(7, 5)$, $(11, 3)$, $(11, 4)$, $(13, 3)$, $(15, 3)$ (`records/bipartite_dd/CANDIDATES.md`), and the 2026-10-20 proofread reports the `geometry.tex` table values as matching the wiki; later sessions could not reach the wiki, and this note did not re-check it (`research/frontier/TABLE.md`). Apart from $(13, 3)$ and $(15, 3)$ we know of no change since 2013 in the cells below.
2. **Corollary C1 needs no Klein correspondence.** `NOTE.md` derived the $(16, 4)$ graph through $Q(4, q)$ and listed the Klein step as not formalised. The same construction is the point–line dual one in $W(3, q)$ itself: one copy of each point of the Singer 2-ovoid and two copies of each line of a regular spread (Section 5). We checked this form by computer for $q = 3, 5$ (`research/theorems/polish/c1w3.py`: orders 120 and 416, diameter 4).
3. **Lean status in `NOTE.md` §7 is out of date.** Theorem B *with* stand-ins is now in Lean (`GQST.standin_gqst`), and so are the regular versions; see Section 9.
4. **`NOTE.md` open problem 3** ($q^2 + 1$ shadow lines on $Q(4, q)$ with one ovoid of clones, q odd) is settled negatively by `geometry.tex`, Proposition 5.13 (at most $q^2 - q + 2$). `NOTE.md` is superseded on this point.
5. **Theorem A in nicer coordinates.** The pair $\kappa_1 = 1/2$, $\kappa_2 = 1/(2\varepsilon)$ satisfies $\kappa_1\kappa_2 = 1/(4\varepsilon)$, so the two quadrics can be written $2X_0X_1 = X_2^2 - \varepsilon X_3^2$ and $2\varepsilon X_0X_1 = X_2^2 - \varepsilon X_3^2$. Same theorem; checked by computer for $q = 3, 5, 7, 9$ (`polish/thmA.py`).
6. **The fourth hat needs $q \geq 3$ without coordinates and $q \geq 4$ in coordinates.** The coordinate-free statement uses four points of a line ($q + 1 \geq 4$); the coordinate version labels them by four field elements $0, 1, a, c$ ($q \geq 4$). Both are stated as such below; no table cell has $q = 3$.
7. **Delorme's two 1985 papers.** Which of [D85a] and [D85b] contains the doubling with levels and the $D = 4, 6$ table constructions is not confirmed (we have not seen the full texts). Both are cited. Tallini's page range (Mitt. Math. Sem. Giessen 201 (1991) 141–147), marked "to be verified" in `geometry.tex`, agrees with independent citing papers and is now given without the caveat. The 2013 values at $\Delta = 16$ (5134 at $D = 4$, 884062 at $D = 6$) are Delorme's doublings with $m = 3$; this is now stated (Example 3.4, Remark 6.9).
8. **Theorem R is explicit except for the completing anti-flags.** The completion step is an existence argument (Lemma 7.2). We now say this plainly: every neighbourhood is listed except the $(m - 1)$

completing anti-flags at the deficient vertices, and neither the maximum degree nor the diameter depends on that choice.

9. **Printed matching in [AL20].** Already noted in `geometry.tex`: with the index set $\mathbb{F} \setminus \{0, 1, a\}$ four vertices keep degree $q + 1$; with $\mathbb{F}^*$ the graph is $(q + 2)$-regular as intended. Kept (Remark 2.7).
10. **Regular fourth hat in arbitrary planes.** `geometry.tex` Remark 4.8 obtained a regular completion in a non-Desarguesian plane only for $q \geq 7$. The count used in the Lean proof (each deficient element is incident with exactly $q - 4$ deficient elements) gives it for every $q \geq 4$ (Section 2.2). The Lean theorem already says so.
11. **Credit to [ADFL21].** `geometry.tex` did not cite Araujo-Pardo–Dalfó–Fiol–López (line doubling in a GQ, the closest prior use of line copies); it does now, and so does Section 3 here.
12. **`geometry.tex` revised to match** (25 October): the 2013 column, the coordinate tables and remarks of this note, the explicit deletion in Theorem R, a Verification section (what Lean covers) and the $(7, 4)$ appendix. Theorem numbers below refer to the revised, recompiled `geometry.pdf`.

Summary of cells

M_b is the bipartite Moore bound. "2013" is the survey table [MS13] (DS14v2, Table 4, as of March 2013); "wiki" is the current CombinatoricsWiki value [CW], shown only where we know it differs from 2013 (see Change 1 for what was checked). All our graphs were rebuilt and checked by the coordinator.

(Δ, D)	M_b	2013	wiki	ours	regular?	result	status
(11,3)	222	190		**192**	yes	fourth hat, $q = 9$	paper, computer, Lean
(13,3)	314	270	274	**276**	yes	fourth hat, $q = 11$	paper, computer, Lean
(15,3)	422	370	374	**376**	yes	fourth hat, $q = 13$	paper, computer, Lean
(7,4)	518	346		**359**	no (350 regular)	Theorem B on $Q(4, 5)$, explicit example	computer, Lean (graph)
(11,4)	2222	1734		**1782**	yes	Theorem A, $q = 9$	paper, computer, Lean
(13,4)	3770	3064		**3146**	yes	Theorem A, $q = 11$	paper, computer, Lean
(15,4)	5910	4946		**5070**	yes	Theorem A, $q = 13$	paper, computer, Lean
(16,4)	7232	5134		**5440**	yes	Prop. C + Cor. C1, $q = 13$	paper, computer, Lean
(7,5)	3110	1134		**1152**	yes	Haar lemma, $S_4 \times \mathbb{Z}_{24}$	paper, computer, Lean
(9,5)	9362	2496		**2662**	yes	Haar lemma, Heisenberg group mod 11	paper, computer, Lean
(7,6)	18662	8992		**9592**	9348	Theorem D / R, $q = 5$	paper, computer, Lean
(11,6)	222222	142464		**148944**	147500	Theorem D / R, $q = 9$	paper, computer, Lean (theorem)
(13,6)	542906	374452		**388972**	386328	Theorem D / R, $q = 11$	paper, computer, Lean (theorem)
(15,6)	1158390	842048		**870440**	866068	Theorem D / R, $q = 13$	paper, computer, Lean (theorem)
(16,6)	1627232	884062		**940846**	927724	Theorem D / R, $q = 13$, $m = 3$	paper, computer, Lean (theorem)

"Lean (theorem)": the general theorem is Lean-checked for every q, so these orders are Lean-checked lower bounds (Section 9); the graphs themselves are too large for a kernel certificate. "Lean (graph)": the specific

graph is kernel-checked. The 2013 values at $D = 4$ and $D = 6$ are Delorme's [D85a, D85b], with $\Delta = q + m$: $m = 2$ at $\Delta = 7, 11, 13, 15$ ($2q^3 + 3q^2 + 3q + 6$ at $D = 4$; $2(q+1)(q^4+q^2+1) + q^4 + 4q^3 + q^2 + 4q + 10$ at $D = 6$) and $m = 3$, $q = 13$ at $\Delta = 16$ (5134 and 884062); see Example 3.4 and Remark 6.9. At $(8, 5)$ the Haar lemma gave $1764 > 1710$ (Lean-checked), since superseded by a 2048-vertex lift that is not covered here.

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0. Conventions

An *incidence structure* $\mathcal{S} = (\mathcal{P}, \mathcal{L}, \mathrm{I})$ has points, lines and a symmetric incidence relation; repeated points and repeated lines (same incidences) are allowed. Its *incidence graph* $\Gamma(\mathcal{S})$ has vertex set $\mathcal{P} \cup \mathcal{L}$ and the flags as edges; it is bipartite with the points on one side. Every bipartite graph with a chosen 2-colouring is of this form, so

$$n_b(\Delta, D) = \max\{|\mathcal{P}| + |\mathcal{L}| : \text{ every element has at most } \Delta \text{ neighbours, } \operatorname{diam}\Gamma(\mathcal{S}) \le D\},$$

$$M_b(\Delta, D) = 2\big(1 + (\Delta - 1) + \cdots + (\Delta - 1)^{D-1}\big).$$

Distances $d(x, y)$ are in $\Gamma(\mathcal{S})$. *Collinear* points are distinct points on a common line; *concurrent* lines are distinct lines on a common point. $\mathbb{F} = \mathrm{GF}(q)$. Points of $\mathrm{PG}(n, q)$ are written $(x_0 : \cdots : x_n)$; lines of $\mathrm{PG}(2, q)$ are written $[u : v : w]$, with $(x : y : z)\,\mathrm{I}\,[u : v : w]$ iff $ux + vy + wz = 0$. χ is the quadratic character of $\mathbb{F}$ (q odd), with $\chi(0) = 0$.

Lemma 0.1 (diameter in geometric terms). Let $\mathcal{S}$ have no isolated element and $D \ge 2$. (i) If D is even, $\operatorname{diam}\mathcal{S} \le D$ iff $d(p, L) \le D - 1$ for every anti-flag (p, L). (ii) If D is odd, $\operatorname{diam}\mathcal{S} \le D$ iff any two points, and any two lines, are at distance at most $D - 1$. In particular: diameter ≤ 3 iff any two points are collinear and any two lines concurrent; diameter ≤ 4 iff for every anti-flag (p, L) some point of L is collinear with p.

Proof. (i) Let $p \ne p'$ be points and L a line on p'. If $p\,\mathrm{I}\,L$, $d(p, p') = 2$; otherwise $d(p, p') \le d(p, L) + 1 \le D$. Lines likewise. (ii) For an anti-flag (p, L) take $p'\,\mathrm{I}\,L$; then $d(p, L) \le d(p, p') + 1 \le D$. The converses hold by parity. □

Lemma 0.2 (adding flags, deleting twins). (a) Adding flags to an incidence structure cannot increase any distance and keeps the incidence graph bipartite. (b) If two lines have the same set of points, deleting one of them changes no distance between the remaining elements (a shortest path through the deleted line can use its twin instead; it does not already contain the twin, since then it could be shortened). □

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1. Doubling with levels, and the hat lemma

Delorme and Farhi [DF84] observed that a *duplicate* of a vertex (a new vertex with the same neighbours) does not increase the diameter. Delorme [D85a, D85b] refined this into a doubling with levels and used it for the old $D = 4$ and $D = 6$ table values. We state it for incidence structures.

1.1 The operation

Let $\mathcal{A} = (\mathcal{A}_0 \supseteq \mathcal{A}_1 \supseteq \cdots \supseteq \mathcal{A}_r \supseteq \mathcal{A}_{r+1} = \emptyset)$ be a filtration of a set $\mathcal{A}_0$ of elements (points and lines may be mixed). The *level* of x is $\lambda(x) = \max\{k : x \in \mathcal{A}_k\}$, and $\lambda(x) = -1$ if $x \notin \mathcal{A}_0$. Fix $m \ge 2$. Two words over $[m] = \{1, \ldots, m\}$ are *comparable* if one is a prefix of the other.

Definition 1.1. The m-fold doubling $\mathcal{S}^{(m)}_{\mathcal{A}}$ has the *versions* x^w, x an element of $\mathcal{S}$ and $w \in [m]^{\lambda(x)+1}$ (a point if x is a point, a line if x is a line), and

$$x^w\ \mathrm{I}\ y^u \iff x\ \mathrm{I}\ y \text{ and } w, u \text{ are comparable.}$$

So an element of level -1 has one version (identified with x), adjacent to *every* version of each of its neighbours; and a version x^w of level $k \ge 0$ is adjacent, for each neighbour y of x, to the single version $y^{w_1 \cdots w_{j+1}}$ if

$\lambda(y) = j \le k$, and to the m^{j-k} versions y^{wu} ($|u| = j-k$) if $\lambda(y) = j > k$. This is the exact neighbourhood of every version.

The map $x^w \mapsto x$ preserves incidence, so $d(x^w, y^u) \ge d(x, y)$ for all versions.

Conditions on $\mathcal{A}$ (with $D \ge 2$):

- **(L0)** every element outside $\mathcal{A}_0$ is incident with at most one element of $\mathcal{A}_0$;
- **(L1)** for every $i \ge 0$: every element incident with an element of $\mathcal{A}_{i+1}$ lies in $\mathcal{A}_i$, and every element of $\mathcal{A}_i \setminus \mathcal{A}_{i+1}$ is incident with at most one element of $\mathcal{A}_{i+1}$;
- **(C_D)** for any two elements x, y (possibly $x = y$) in the same connected component of the substructure induced on $\mathcal{A}_0$, there is $z \notin \mathcal{A}_0$ with $d(x, z) + d(z, y) \le D$.

1.2 Delorme's theorem

Theorem 1.2 (Delorme [D85a, D85b]). Let $\mathcal{S}$ have diameter at most $D \ge 2$ and let $m \ge 2$.

- (a) If (C_D) holds, $\mathcal{S}_{\mathcal{A}}^{(m)}$ has diameter at most D.
- (b) If (L0) and (L1) hold, every version satisfies $\deg x^w \le \deg x + m - 1$, with equality only if x is incident with an element of level $\lambda(x) + 1$.
- (c) $\mathcal{S}_{\mathcal{A}}^{(m)}$ has $|\mathcal{P}| + (m-1)\sum_{k\ge 0} m^k |\mathcal{A}_k \cap \mathcal{P}|$ points and $|\mathcal{L}| + (m-1)\sum_{k\ge 0} m^k |\mathcal{A}_k \cap \mathcal{L}|$ lines.
- (d) Conversely, if $\mathcal{S}_{\mathcal{A}}^{(m)}$ has diameter at most D, the conclusion of (C_D) holds for every pair $x, y \in \mathcal{A}_0$.

Proof. (c) An element of level k has $m^{k+1} = 1 + (m-1)(1 + m + \cdots + m^k)$ versions.

(b) Let $\lambda(x) = k$. By the description after Definition 1.1, only neighbours of level $> k$ contribute more than one version. If $k = -1$, (L0) gives at most one neighbour in $\mathcal{A}_0$, of level exactly 0 (a neighbour in $\mathcal{A}_1$ would put x in $\mathcal{A}_0$ by (L1)); it contributes m versions. If $k \ge 0$, (L1) with $i = k$ gives at most one neighbour in $\mathcal{A}_{k+1}$, and (L1) with $i = k+1$ gives none in $\mathcal{A}_{k+2}$; that neighbour contributes m versions. So $\deg x^w \le \deg x - 1 + m$.

(c) For an infinite word ω the *sheet* Σ_ω is the set of versions x^w with w a prefix of ω. It contains one version of each element, and $x^w \mapsto x$ is an isomorphism from the substructure on Σ_ω onto $\mathcal{S}$. Two versions with comparable words lie in a common sheet, hence are at distance $\le D$. Let x^w, y^u have incomparable words; both are non-empty, so $x, y \in \mathcal{A}_0$. If x, y are in different components of the substructure on $\mathcal{A}_0$, a shortest path $x = z_0, \ldots, z_k = y$ ($k \le D$) passes through some $z_i \notin \mathcal{A}_0$, whose unique version lies in every sheet. Lift $z_0, \ldots, z_i$ into a sheet through x^w and $z_i, \ldots, z_k$ into a sheet through y^u: a path of length k. If x, y are in the same component (including $x = y$), (C_D) gives $z \notin \mathcal{A}_0$ with $d(x, z) + d(z, y) \le D$, and the same lifting gives a path of length $\le D$.

(d) Take versions x^w, y^u with w starting with the letter 1 and u with the letter 2, and a path of length $\le D$ between them. Consecutive versions on it have comparable words, and comparable non-empty words start with the same letter; so some version on the path has the empty word, i.e. is an element $z \notin \mathcal{A}_0$. Projecting the path gives a walk $x \to z \to y$ of length $\le D$. □

1.3 The hat lemma

A *generalized n-gon* is an incidence structure whose incidence graph has diameter n and girth $2n$; it has order (s, t) if lines have $s + 1$ points and points are on $t + 1$ lines.

Lemma 1.3 (hat lemma). Let $\mathcal{S}$ be a generalized n-gon of order (s, t), $s, t \ge 1$, and $\mathcal{A}_0$ a set of elements. Then (C_n) holds iff every connected component of the substructure induced on $\mathcal{A}_0$ is a tree of diameter at most $n - 2$.

Proof. Suppose a component B contains a path $x = z_0, \ldots, z_{n-1} = y$ of length $n - 1$ (it does if B has diameter $\ge n - 1$, and if B contains a cycle, since cycles have length $\ge 2n$). By the girth this is the only path of length $\le n - 1$ from x to y, so $d(x, y) = n - 1$. If $d(x, z) + d(z, y) \le n$ then, by parity, $d(x, z) + d(z, y) = n - 1$, so z lies on a shortest x–y path, i.e. $z = z_i \in \mathcal{A}_0$. So ($\mathrm{C}_n$) fails.

Conversely let B be a tree of diameter $\le n-2$, $x, y \in B$, and P the path from x to y in B, of length d. $\Gamma(\mathcal{S})$ is connected and has cycles, so $B \neq \Gamma(\mathcal{S})$ and some element of B has a neighbour outside $\mathcal{A}_0$. Let k be the least distance in B from P to such an element u, and z a neighbour of u outside $\mathcal{A}_0$. Then $d(x,z)+d(z,y) \le d+2k+2$. If $k=0$ this is $\le n$. If $k \ge 1$, every element of B at distance $< k$ from P has all its neighbours in B, and every element has at least two neighbours ($s, t \ge 1$). So we can walk k steps away from P starting at x, and k steps away from P starting at y (along two different neighbours if $x = y$); in the tree B this gives a path of length $d+2k$. Hence $d+2k \le n-2$ and $d(x,z)+d(z,y) \le n$. □

Corollary 1.4. In a generalized n-gon of order (s,t), $s,t \ge 1$, and for any filtration and $m \ge 2$: $\mathcal{S}_{\mathcal{A}}^{(m)}$ has diameter at most n iff every component of the substructure on $\mathcal{A}_0$ is a tree of diameter at most $n-2$. (Theorem 1.2(a),(d) and Lemma 1.3.) For $n=3$ the allowed components are single elements and flags; for $n=4$ also *stars* (a point with some lines on it, or a line with some of its points); for $n=6$ trees of diameter ≤ 4.

Proposition 1.5 (diameter 3 needs patches). In a projective plane, if $\mathcal{A}$ satisfies (L0), (L1) and (C_3), then $\mathcal{A}_0$ has at most one point and at most one line, and $\mathcal{A}_1 = \emptyset$.

Proof. If $x \neq y$ are points of $\mathcal{A}_0$, the line xy is either outside $\mathcal{A}_0$ and incident with two elements of $\mathcal{A}_0$ (against (L0)), or in $\mathcal{A}_0$, and then x, xy, y is a path of length $2 > n-2$ in one component (against Lemma 1.3). Lines likewise. An element of $\mathcal{A}_1$ would have its $q+1 \ge 3$ neighbours in $\mathcal{A}_0$ by (L1). □

So pure doubling gains at most $2(m-1)$ elements in a projective plane (2 for $m=2$); the hats of Section 2 double flags and then *patch* a bounded number of incidences.

Status. Paper proof (above; `geometry.tex` §3). Lean-checked, for any finite graph: Theorem 1.2(a)–(c) under (L0), (L1) and (C_D) together, with the degree bound $\Delta + m - 1$ (the "equality only if" clause of (b) and part (d) are not formalised) (`Doubling.delorme_levelled`, `isBipartiteDegDiam_levelled`, `card_levelled_levels`), the hat lemma in both directions for generalized polygons given as graphs of diameter n and girth $2n$ with all degrees ≥ 2 (`Doubling.GenPolygonGraph.condC_iff_forestDiam`), and Corollary 1.4's "if" direction (`delorme_levelled_polygon`).

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2. The fourth hat (diameter 3)

Araujo-Pardo and López [AL20] add three *hats* to $\mathrm{PG}(2,q)$ and obtain $n_b(q+2,3) \ge 2(q^2+q+5)$. We add a fourth.

2.1 Frame and notation

$\ell = [1:0:0]$ (the line $x=0$), $A = (1:0:0) \notin \ell$. The points of ℓ and the lines on A are

$$X_t = (0:t:1),\ P = (0:1:0); \qquad \lambda_t = [0:1:-t],\ \lambda_\infty = [0:0:1] \qquad (t \in \mathbb{F}).$$

Every point is listed once among A, X_t ($t \in \mathbb{F}$), P, $(x:y:1)$ ($x \neq 0$), $(x:1:0)$ ($x \neq 0$); every line once among ℓ, λ_t, λ_∞, $[u:1:w]$ ($u \neq 0$), $[1:0:w]$ ($w \neq 0$). Facts used (one-line computations):

- (F1) $\lambda_t \cap \ell = X_t$, $\lambda_\infty \cap \ell = P$; the points of λ_t are A and $(x:t:1)$, $x \in \mathbb{F}$ (X_t at $x=0$); those of λ_∞ are A and $(x:1:0)$.
- (F2) The lines on X_t are ℓ and $[w:1:-t]$, $w \in \mathbb{F}$ (λ_t at $w=0$); the lines on P are ℓ, λ_∞ and $[1:0:w]$, $w \neq 0$.
- (F3) If $u = (x:y:z)$ with $x \neq 0$, then u and X_s lie on $[(sz-y)/x : 1 : -s] \neq \ell$.
- (F4) A line $[u:v:w]$ with $u \neq 0$ meets λ_s in $(-(vs+w)/u : s : 1) \neq A$; $[u:1:w]$ meets ℓ in X_{-w}.

Let $q \ge 4$, $a, c \in \mathbb{F} \setminus \{0,1\}$ distinct, and

$$S = \{0,1,a,c\}, \qquad T = \{0,1,c\}, \qquad \sigma = (0\ c\ 1) :\ 0 \mapsto c \mapsto 1 \mapsto 0.$$

2.2 Statement

Definition 2.1. $\mathcal{S}(q;a,c)$ has the points of $\mathrm{PG}(2,q)$ and five new points Q, H_0, H_1, H_a, H_c, the lines of $\mathrm{PG}(2,q)$ and five new lines M, K_0, K_1, K_a, K_c, and exactly these incidences:

- (C1) every incidence of $\mathrm{PG}(2,q)$ except $X_s \,\mathrm{I}\, \lambda_s$ ($s \in S$), plus the *rotation* $X_0 \,\mathrm{I}\, \lambda_c$, $X_c \,\mathrm{I}\, \lambda_1$, $X_1 \,\mathrm{I}\, \lambda_0$ (that is $X_t \,\mathrm{I}\, \lambda_{\sigma(t)}$, $t \in T$);
- (C2) H_s lies on $[w:1:-s]$ for every $w \in \mathbb{F}$ ($s \in S$): the lines on X_s other than ℓ;
- (C3) K_s contains $(x:s:1)$ for every $x \in \mathbb{F}$ ($s \in S$): the points of λ_s other than A;
- (C4) M contains P, X_t ($t \in \mathbb{F} \setminus T$), and H_0, H_1, H_a, H_c;
- (C5) Q lies on λ_∞, λ_t ($t \in \mathbb{F} \setminus T$), and K_0, K_1, K_a, K_c;
- (C6) the *links* $A \,\mathrm{I}\, K_a$, $H_a \,\mathrm{I}\, \ell$, $H_0 \,\mathrm{I}\, K_c$, $H_c \,\mathrm{I}\, K_1$, $H_1 \,\mathrm{I}\, K_0$ (that is $H_t \,\mathrm{I}\, K_{\sigma(t)}$, $t \in T$).

$G(q;a,c)$ is its incidence graph. $R(q;a,c)$ is $G(q;a,c)$ plus the *filler* edges

$$(x:y:1) \sim [x:1:-y] \quad (x \neq 0,\ y \notin S), \qquad (x:1:0) \sim [1:0:-1/x] \quad (x \neq 0).$$

Theorem 2.2. For every prime power $q \geq 4$ and distinct $a, c \in \mathbb{F} \setminus \{0,1\}$, $G(q;a,c)$ and $R(q;a,c)$ are bipartite, have $2(q^2+q+6)$ vertices, maximum degree $q+2$ and diameter 3, and $R(q;a,c)$ is $(q+2)$-regular. Hence $n_b(q+2,3) \geq 2(q^2+q+6)$, attained by a regular graph.

The construction and proof use only the axioms of a projective plane together with an anti-flag (A,ℓ) and four points X_s of ℓ, so the coordinate-free version holds in every projective plane of order $q \geq 3$ (`geometry.tex`, Theorem 4.2), and so does a regular completion, for every $q \geq 4$: a deficient point p lies on exactly $q-4$ deficient lines (the lines on p other than Ap meet ℓ in the q points other than $Ap \cap \ell$, four of which are the X_s), dually a deficient line has exactly $q-4$ deficient points, and $2(q-4) \leq (q-1)(q-3)$, so Hall's theorem gives a perfect matching of non-incident deficient pairs (this is the argument of the Lean proof; `geometry.tex` Remark 4.8 had the weaker bound $q+1$ and so needed $q \geq 7$). The hat pair (H_s, K_s) is a copy of the flag (X_s, λ_s) with the flag "crossed", and (Q, M) is a copy of the anti-flag (A, ℓ).

2.3 New vertices and every changed neighbourhood

vertex	neighbours in $G(q;a,c)$	degree
Q	λ_∞, λ_t ($t \notin T$), K_0, K_1, K_a, K_c	$q+2$
H_a	$[w:1:-a]$ ($w \in \mathbb{F}$), M, ℓ	$q+2$
H_t, $t \in T$	$[w:1:-t]$ ($w \in \mathbb{F}$), M, $K_{\sigma(t)}$	$q+2$
M	P, X_t ($t \notin T$), H_0, H_1, H_a, H_c	$q+2$
K_a	$(x:a:1)$ ($x \in \mathbb{F}$), Q, A	$q+2$
K_t, $t \in T$	$(x:t:1)$ ($x \in \mathbb{F}$), Q, $H_{\sigma^{-1}(t)}$	$q+2$
A	λ_t ($t \in \mathbb{F}$), λ_∞, K_a	$q+2$
P	ℓ, λ_∞, $[1:0:w]$ ($w \neq 0$), M	$q+2$
X_t, $t \notin S$	ℓ, $[w:1:-t]$ ($w \in \mathbb{F}$), M	$q+2$
X_a	ℓ, $[w:1:-a]$ ($w \neq 0$), M, K_a	$q+2$
X_t, $t \in T$	ℓ, $[w:1:-t]$ ($w \neq 0$), $\lambda_{\sigma(t)}$, K_t	$q+2$
$(x:y:1)$, $x \neq 0$	$[u:1:-(ux+y)]$ ($u \in \mathbb{F}$), $[1:0:-x]$, and K_y if $y \in S$	$q+2$ if $y \in S$, else $q+1$
$(x:1:0)$, $x \neq 0$	λ_∞, $[1:-x:w]$ ($w \in \mathbb{F}$)	$q+1$
ℓ	P, X_t ($t \in \mathbb{F}$), H_a	$q+2$
λ_∞	A, $(x:1:0)$ ($x \in \mathbb{F}$), Q	$q+2$
λ_t, $t \notin S$	A, $(x:t:1)$ ($x \in \mathbb{F}$), Q	$q+2$
λ_a	A, $(x:a:1)$ ($x \neq 0$), Q, H_a	$q+2$
λ_t, $t \in T$	A, $(x:t:1)$ ($x \neq 0$), $X_{\sigma^{-1}(t)}$, H_t	$q+2$
$[u:1:w]$, $u \neq 0$	$(x:-ux-w:1)$ ($x \in \mathbb{F}$), $(-1/u:1:0)$, and H_{-w} if $-w \in S$	$q+2$ if $-w \in S$, else $q+1$
$[1:0:w]$, $w \neq 0$	$(-w:y:1)$ ($y \in \mathbb{F}$), P	$q+1$

Here $\sigma^{-1}(0) = 1$, $\sigma^{-1}(1) = c$, $\sigma^{-1}(c) = 0$. This is the neighbourhood of *every* vertex. The vertices of degree $q+1$ are the points $(x : y : 1)$ $(x \neq 0,\ y \notin S)$, $(x : 1 : 0)$ $(x \neq 0)$, and the lines $[u : 1 : w]$ $(u \neq 0,\ -w \notin S)$, $[1 : 0 : w]$ $(w \neq 0)$: $(q-1)(q-3)$ of each. The filler adds exactly one neighbour to each of them.

2.4 Proof of Theorem 2.2

Degrees (Lemma 2.3). Each row of the table is (F1) or (F2) corrected by (C1)–(C6); within a row the listed neighbours are distinct. For instance: A keeps its $q+1$ lines and gains only K_a (A is not of the form $(x : s : 1)$); X_a loses λ_a and gains M (as $a \notin T$) and K_a; X_t $(t \in T)$ loses λ_t and gains $\lambda_{\sigma(t)}$ and K_t, and is not on M; a point $(x : y : 1)$ with $x \neq 0$ is not on ℓ, so it loses nothing, and it gains K_s iff $y = s \in S$; Q is on $1 + (q-3) + 4 = q+2$ lines; H_s is on q lines by (C2), on M, and on exactly one link line. The line rows are the dual count: ℓ gains only H_a; λ_a loses X_a and gains Q, H_a; λ_t $(t \in T)$ loses X_t, gains $X_{\sigma^{-1}(t)}$ and H_t; $[u : 1 : w]$ $(u \neq 0)$ gains H_s iff $s = -w \in S$; M has $1 + (q-3) + 4$ points; K_s has q points, Q, and one link point. The filler map $\delta : (x : y : z) \mapsto [x : z : -y]$ is a correlation of $\mathrm{PG}(2, q)$ $(ux + wz + (-v)(-y) = ux + vy + wz)$, a point lies on its image iff $x^2 = 0$, i.e. iff it is on ℓ, and δ maps $(x : y : 1) \mapsto [x : 1 : -y]$ and $(x : 1 : 0) \mapsto [x : 0 : -1] = [1 : 0 : -1/x]$. So it is a bijection from the deficient points to the deficient lines, and the filler edges are new (non-incident pairs). Hence G has maximum degree $q+2$ and R is $(q+2)$-regular.

Reduction (Lemma 2.4). If every line has a point, any two points are collinear and any two lines concurrent, the incidence graph has diameter ≤ 3: for an anti-flag (p, L) take $p' \,\mathrm{I}\, L$ and a line L' on p, p'; then p, L', p', L is a path. (Lemma 0.1(ii).)

In the next two lemmas "incident" means incident in $\mathcal{S}(q; a, c)$. The only incidences of $\mathrm{PG}(2, q)$ missing are $X_s \,\mathrm{I}\, \lambda_s$, $s \in S$; so a plane incidence $u \,\mathrm{I}\, L$ survives whenever $u \notin \ell$ or L is not on A.

Points (Lemma 2.5). Any two points $u \neq v$ are collinear.

- (P1) Both in $\mathrm{PG}(2, q)$, $L = uv$. If both incidences survive, use L. Otherwise, say $u = X_s$, $L = \lambda_s$, $s \in S$, and $v = A$ or $v = (x : s : 1)$, $x \neq 0$. If $v = (x : s : 1)$: K_s. If $v = A$, $s = a$: K_a (C3, C6). If $v = A$, $s \in T$: $\lambda_{\sigma(s)}$ (rotation).
- (P2) Q and u. $u \in \{A, P\}$ or $u = (x : 1 : 0)$: λ_∞. $u = X_t$, $t \notin S$: λ_t. $u = X_s$, $s \in S$: K_s. $u = (x : y : 1)$, $x \neq 0$: λ_y if $y \notin T$, K_y if $y \in T$.
- (P3) H_s and u. $u \notin \ell$ (including A): the line of (F3), which contains H_s by (C2). $u = X_s$: $[1 : 1 : -s]$. $u = P$ or $u = X_t$, $t \notin T$, $t \neq s$: M. $u = X_t$, $t \in T \setminus \{s\}$, $s = a$: ℓ. $u = X_t$, $t \in T \setminus \{s\}$, $s \in T$: as σ is a 3-cycle, $t = \sigma(s)$ or $t = \sigma^{-1}(s)$; use $K_{\sigma(s)}$ (contains $X_{\sigma(s)}$ and H_s) or λ_s (contains $X_{\sigma^{-1}(s)}$ by the rotation, and H_s by (C2) with $w = 0$).
- (P4) Two new points. $H_s, H_{s'}$: M. Q, H_a: λ_a. Q, H_t $(t \in T)$: $K_{\sigma(t)}$.

Lines (Lemma 2.6). Any two lines $L \neq N$ are concurrent.

- (N1) Both in $\mathrm{PG}(2, q)$, $p = L \cap N$. If both incidences survive, use p. Otherwise, say $p = X_s$, $L = \lambda_s$, and $N = \ell$ or $N = [w : 1 : -s]$, $w \neq 0$. If $N = [w : 1 : -s]$: H_s. If $N = \ell$, $s = a$: H_a. If $N = \ell$, $s \in T$: $X_{\sigma^{-1}(s)}$ (rotation).
- (N2) M and L. L on P ($L = \ell, \lambda_\infty$ or $[1 : 0 : w]$): P. $L = \lambda_t$, $t \notin S$: X_t. $L = \lambda_s$, $s \in S$: H_s. $L = [u : 1 : w]$, $u \neq 0$: it meets ℓ in X_r, $r = -w$ (F4); use X_r if $r \notin T$, and H_r if $r \in T$ (H_r is on $[u : 1 : -r]$ and on M).
- (N3) K_s and L. L not on A ($L = \ell$ or $[u : v : w]$, $u \neq 0$): $L \cap \lambda_s = (x : s : 1)$ (F1, F4), which is on K_s, and its incidence with L survives. $L = \lambda_s$: $(1 : s : 1)$. $L = \lambda_\infty$ or λ_t, $t \notin T$: Q. $L = \lambda_t$, $t \in T \setminus \{s\}$, $s = a$: A. $L = \lambda_t$, $t \in T \setminus \{s\}$, $s \in T$: if $t = \sigma(s)$, X_s (on $\lambda_{\sigma(s)}$ and K_s); if $t = \sigma^{-1}(s)$, H_t (on λ_t and on $K_{\sigma(t)} = K_s$).
- (N4) Two new lines. $K_s, K_{s'}$: Q. M, K_a: X_a. M, K_t $(t \in T)$: $H_{\sigma^{-1}(t)}$.

Conclusion. By Lemmas 2.4–2.6, G has diameter ≤ 3; adding the filler cannot increase it (Lemma 0.2). The graphs are bipartite and not complete bipartite ($A \notin \ell$), so the diameter is exactly 3. □

(The proof uses only that $\mathbb{F}$ is a field and $0, 1, a, c$ are distinct. The correlation δ, extended by $H_s \leftrightarrow K_s$, $Q \leftrightarrow M$, maps $\mathcal{S}(q; a, c)$ with σ onto the dual of the structure built with σ^{-1}, so Lemma 2.6 is also Lemma 2.5 for σ^{-1}; we wrote both out so that every case can be checked directly.)

Remark 2.7 (the three-hat structure of [AL20] in these coordinates). Hats $S_{AL} = \{0, 1, a\}$, no rotation, only the flag (X_a, λ_a) removed, links $A \,\mathrm{I}\, K_a$, $H_a \,\mathrm{I}\, \ell$, $H_0 \,\mathrm{I}\, K_1$, $H_1 \,\mathrm{I}\, K_0$; M contains P and X_t $(t \neq 0, 1)$; Q lies on λ_∞ and λ_t $(t \neq 0, 1)$. Dictionary with the affine coordinates of [AL20]: $(x, y) = (x : y : 1)$, $[m, b] = [m : -1 : b]$,

$L_c = [1:0:-c]$, $L = [0:0:1]$, $P_m = (1:m:0)$; so $L_0 = \ell$, $P_0 = A$, $(0,s) = X_s$, $[0,s] = \lambda_s$, $L = \lambda_\infty$. The completing matching printed in [AL20] is $(x,y) \sim [x,y]$ ($x \neq 0$, $y \notin S_{AL}$) and $P_i \sim L_i$ for $i \in \mathbb{F} \setminus \{0,1,a\}$; with this index set P_1, P_a, L_1, L_a keep degree $q+1$. With $i \in \mathbb{F}^*$ the graph is $(q+2)$-regular, as intended. This does not affect the maximum degree or the diameter.

Cells. $q = 9, 11, 13$ give $(11,3)$: 192 (2013: 190), $(13,3)$: 276 (2013: 270; wiki 274 [AL20]), $(15,3)$: 376 (2013: 370; wiki 374 [AL20]), all regular. At $q = 5$ it gives $72 < 80 = n_b(7,3)$ (optimal value known).

Status. Paper proof (above; `coords.tex`; coordinate-free in `geometry.tex` §4). Computer-checked: `research/plus2/coords/coords.py` builds the structure three ways (from (C1)–(C6), from the point rows and from the line rows of the table), checks they agree, and checks degrees, both covering properties and the exact diameter of G and R for every prime power $q \leq 13$ and every admissible (a,c); removing the rotation or the link $H_1 \,\mathrm{I}\, K_0$ gives diameter 4 in every case tested. Lean-checked: the regular version for every finite projective plane of order $q \geq 4$ with a chosen frame (`Plus2.plus2_regular`, `plus2_regular_of_order`), and the three record graphs (`nb_11_3`, `nb_13_3`, `nb_15_3`); see Section 9.

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3. Theorem B: clones and shadow lines in a generalized quadrangle

A *generalized quadrangle* (GQ) of order (s,t), $s,t \geq 1$, has $(s+1)(st+1)$ points and $(t+1)(st+1)$ lines, and for every anti-flag (p,L) exactly one point of L is collinear with p. Its incidence graph has diameter 4. Copies of lines in a GQ appear already in Araujo-Pardo, Dalfó, Fiol and López [ADFL21, Sec. 4], who double every line to get biregular graphs (semi-double graphs). Theorem B copies only some lines, with weights, lets a copy attach to clones (stand-ins), and combines this with weighted point copies in the spirit of Delorme–Farhi duplication [DF84].

3.1 Statement

Definition 3.1 (clones and shadows). Let $\mathcal{S} = (\mathcal{P}, \mathcal{L})$ be a GQ of order (s,t), let $w : \mathcal{P} \to \mathbb{Z}_{\geq 0}$ be a weight, and $\mathcal{M}$ a finite multiset of lines. For each entry μ of $\mathcal{M}$, with underlying line L_μ, choose for every point $p \,\mathrm{I}\, L_\mu$ a *stand-in* $\sigma_\mu(p) \in \{p, p^1, \ldots, p^{w(p)}\}$. The structure $\mathcal{S}[w, \mathcal{M}, \sigma]$ has

- points: $\mathcal{P}$ and the *clones* $p^1, \ldots, p^{w(p)}$ of each $p \in \mathcal{P}$;
- lines: $\mathcal{L}$ and one *shadow line* L^μ for each entry μ of $\mathcal{M}$;
- incidences: those of $\mathcal{S}$; $p^i \,\mathrm{I}\, L$ whenever $p \,\mathrm{I}\, L \in \mathcal{L}$; and $\sigma_\mu(p) \,\mathrm{I}\, L^\mu$ for $p \,\mathrm{I}\, L_\mu$.

New vertices and changed neighbourhoods.

vertex	neighbours	degree
clone p^i	the $t+1$ old lines on p; the shadow lines L^μ with $\sigma_\mu(p) = p^i$	$t + 1 + \#\{\mu : \sigma_\mu(p) = p^i\}$
shadow line L^μ	$\sigma_\mu(p)$ for the $s+1$ points p of L_μ	$s+1$
old point p	its $t+1$ lines; the shadow lines L^μ with $\sigma_\mu(p) = p$	$t + 1 + \#\{\mu : \sigma_\mu(p) = p\}$
old line L	its $s+1$ points and all their clones	$s + 1 + \sum_{p \,\mathrm{I}\, L} w(p)$

When all stand-ins are old points ($\sigma = \mathrm{id}$), a shadow line is adjacent exactly to the old points of L_μ, and clones are adjacent only to old lines.

Theorem 3.2 (Theorem B). Let $\mathcal{S}$ be a GQ of order (s,t), $s,t \geq 1$, and $\mathcal{S}' = \mathcal{S}[w, \mathcal{M}, \sigma]$.

- (a) $\mathcal{S}'$ has diameter 4.
- (b) $\mathcal{S}'$ has maximum degree $\leq \Delta$ iff $\Delta \geq \max(s,t) + 1$, $\sum_{p \,\mathrm{I}\, L} w(p) \leq \Delta - s - 1$ for every line L of $\mathcal{S}$, and every point of $\mathcal{S}'$ (old point or clone) is a stand-in for at most $\Delta - t - 1$ entries of $\mathcal{M}$.
- (c) $\mathcal{S}'$ has $(s+1)(st+1) + \sum_p w(p)$ points and $(t+1)(st+1) + |\mathcal{M}|$ lines.

3.2 Proof

(b) and (c) are read off the table. (a) Write π for the map sending a clone or shadow line to its underlying point or line. Every point x of $\mathcal{S}'$ is incident with every old line through $\pi(x)$. We use Lemma 0.1(i); no element is isolated. Let (x, Y) be an anti-flag of $\mathcal{S}'$.

- Y old. Then $\pi(x) \notin Y$ (else $x \,\mathrm{I}\, Y$). Let r be the point of Y collinear with $\pi(x)$, on the line N. Then $x \,\mathrm{I}\, N \,\mathrm{I}\, r \,\mathrm{I}\, Y$.
- $Y = L^\mu$, $\pi(x) \notin L_\mu$. Let $r \,\mathrm{I}\, L_\mu$ be collinear with $\pi(x)$, on the line N. Then $x \,\mathrm{I}\, N \,\mathrm{I}\, \sigma_\mu(r) \,\mathrm{I}\, L^\mu$.
- $Y = L^\mu$, $\pi(x) \,\mathrm{I}\, L_\mu$. Pick $r \,\mathrm{I}\, L_\mu$, $r \neq \pi(x)$ ($s \geq 1$). Then $x \,\mathrm{I}\, L_\mu \,\mathrm{I}\, \sigma_\mu(r) \,\mathrm{I}\, L^\mu$.

So $d(x, Y) \leq 3$ for every anti-flag, and the diameter is ≤ 4. Two non-collinear old points (they exist, since a point is collinear with $s(t+1) < (s+1)(st+1) - 1$ others) stay non-collinear: they share no old line, a shadow line L^μ contains only points over points of L_μ, and an old point lies over itself only. So the diameter is 4. □

Corollary 3.3 (exact weights). Let $\Delta \geq \max(s, t) + 1$, and suppose w is a weighted $(\Delta - s - 1)$-ovoid ($\sum_{p \,\mathrm{I}\, L} w(p) = \Delta - s - 1$ for every line) and $\mathcal{M}$ a weighted $(\Delta - t - 1)$-cover (every point on exactly $\Delta - t - 1$ entries). Then $\mathcal{S}[w, \mathcal{M}, \mathrm{id}]$ has diameter 4, maximum degree Δ and $2\Delta(st+1)$ elements. It becomes Δ-regular, with the same order and diameter 4, by the following explicit flags: number the clones $c_0, c_1, \ldots$ and the shadow lines $y_0, \ldots, y_{b-1}$ ($b = |\mathcal{M}|$), put $k = \Delta - t - 1$, and join c_i to y_j for $j \equiv ik, ik+1, \ldots, ik+k-1 \pmod b$.

Proof. Double counting flags gives $\sum_p w(p) = (\Delta - s - 1)(st+1)$ and $|\mathcal{M}| = (\Delta - t - 1)(st+1)$; the total is $(s + t + 2 + \Delta - s - 1 + \Delta - t - 1)(st+1) = 2\Delta(st+1)$. Old points and lines have degree Δ, clones $t+1$, shadow lines $s+1$. The added pairs are anti-flags (with $\sigma = \mathrm{id}$ no clone is on a shadow line); the blocks $\{ik, \ldots, ik+k-1\}$, $i = 0, 1, \ldots$, run through $0, 1, \ldots, k\sum_p w(p) - 1 = b(\Delta - s - 1) - 1$, so modulo b every residue is hit exactly $\Delta - s - 1$ times, and within a block the residues are distinct as $k \leq b$. Adding flags does not increase distances, and two non-collinear old points still share no line. □

Example 3.4 (Delorme's $D = 4$ construction [D85a, D85b], the 2013 values). In $W(q)$ take a spread Σ, a line $v \in \Sigma$, and $\mathcal{A}_1 = \{v\}$, $\mathcal{A}_0 = \Sigma \cup \{\text{points of } v\}$. (L1): the elements on v are its points; a point of v meets only v in $\mathcal{A}_1$, a line of $\Sigma \setminus \{v\}$ nothing. (L0): a point off v is on exactly one line of Σ and on no other line of $\mathcal{A}_0$; a line not in Σ has at most one point on v. The components are the star $v \cup \{\text{points of } v\}$ and single lines, so (C_4) holds (Lemma 1.3). Theorem 1.2 gives maximum degree $q + m$, diameter 4 and $2(q+1)(q^2+1) + (m-1)\big((q^2+1) + (q+1) + m\big)$ elements. For $m = 2$ this is $2q^3 + 3q^2 + 3q + 6$: 346, 1734, 3064, 4946 at $q = 5, 9, 11, 13$; for $m = 3$, $q = 13$ it is $4760 + 2 \cdot 187 = 5134$, the 2013 value at $\Delta = 16$.

Remark 3.5 (limit of shadow lines on $Q(4, q)$, `geometry.tex` Prop. 5.13). For q odd, an ovoid $\mathcal{O}$ of $Q(4, q)$, $w = \mathbf{1}_\mathcal{O}$ and maximum degree $q + 2$, at most $q^2 - q + 2$ shadow lines are possible: every old point and clone serves at most one shadow line, so a point off $\mathcal{O}$ is on at most one entry of $\mathcal{M}$ and a point of $\mathcal{O}$ on at most two; if some $o \in \mathcal{O}$ is on two entries, each other entry L misses o and so has a unique point y collinear with o; $y \notin \mathcal{O}$ (points of an ovoid are pairwise non-collinear); the line oy is not an entry of $\mathcal{M}$ (else $y \notin \mathcal{O}$ would be on two entries), so it is one of the $q - 1$ lines on o other than the two entries through o, and y is one of its q points other than o; distinct L give distinct y. So $|\mathcal{M}| \leq 2 + q(q-1)$. Otherwise $\mathcal{M}$ is a partial spread and $|\mathcal{M}| \leq q^2 - q + 1$ (Tallini [Tal91]; see also [CDPS22], §1).

Application: $(7, 4) \geq 359$. $Q(4, 5)$, one clone of each point of an elliptic-quadric ovoid ($w = \mathbf{1}_\mathcal{O}$), and 21 shadow lines with stand-ins: $2 \cdot 6 \cdot 26 + 26 + 21 = 359$, maximum degree 7. Such examples come from an integer program (HiGHS proves 21 optimal for this family; `research/gq4x/q4_share_ilp.py`, earlier optimum in `research/gq4x/graphs/q4_share_q5_n359.json`); they are not members of a proved infinite family. The record file `records/bipartite_dd/d7D4/359-spec-d7d4-g017-c11.json` (found by evolutionary search) decomposes the same way after deleting 21 redundant edges, with stand-ins essential. Every ovoid of $Q(4, p)$, p prime, is an elliptic quadric [BGS06], so the choice of ovoid does not matter. Appendix A lists, in coordinates, an optimum of the same integer program re-solved for this note (a different set of 21 lines from the earlier file), so that such a graph can be checked by hand. Cell: $(7, 4)$, 2013: 346, ours 359 (regular: 350 from Theorem A at $q = 5$).

Status. Paper proof (above; `NOTE.md` §2, `geometry.tex` Theorem 5.4). Computer-checked on many instances (`research/gq4`, `research/gq4x`). Lean-checked for every GQ of order (s, t): the version with all stand-ins original and arbitrary point and line multiplicities (`GQST.gqst`), the version with stand-ins (`GQST.standin_gqst`), and the regular completion (`GQST.regular_copy_graph`); the 359 graph itself (`nb_7_4`). See Section 9.

4. Theorem A: the Ahrens–Szekeres quadrangle and two elliptic quadrics

4.1 Coordinates

Let q be odd. $W(q) = W(3,q)$ is the symplectic quadrangle of $\mathrm{PG}(3,q)$ for the alternating form $\beta(u,v) = u_0v_1 - u_1v_0 + u_2v_3 - u_3v_2$. Let $x = (1:0:0:0)$; then $x^\perp$ is the plane $X_1 = 0$. Payne's derived quadrangle $\mathrm{AS}(q) = P(W(q), x)$ [P71], [PT, 3.1.4] has

- **points**: the q^3 points $(a:1:b:c)$, $a,b,c \in \mathbb{F}$ (the points off $x^\perp$); write (a,w) with $w = (b,c)$;
- **lines of type (i)** (*fibres*), q^2 of them: $\{(a_0+\mu : 1 : w_0) : \mu \in \mathbb{F}\}$, the line $\langle x, (a_0:1:w_0)\rangle$ minus x;
- **lines of type (ii)**, $q^2(q+1)$ of them: for a point $(a_0:1:w_0)$ and a direction $d = (d_1:d_2) \in \mathrm{PG}(1,q)$,

$$\ell(a_0, w_0; d) = \{(a_0 + \lambda\rho : 1 : w_0 + \lambda d) : \lambda \in \mathbb{F}\}, \qquad \rho = \det(w_0, d) = b_0d_2 - c_0d_1,$$

  the totally isotropic line through $(a_0:1:w_0)$ and $(\rho:0:d_1:d_2)$, minus the latter point.

(Check: $\beta((a_0,1,b_0,c_0),(\rho,0,d_1,d_2)) = -\rho + b_0d_2 - c_0d_1 = 0$. A point off $x^\perp$ is on $q+1$ totally isotropic lines, none through x, each meeting $x^\perp$ in one point $(\rho:0:d)$, $d \neq 0$; so each point of $\mathrm{AS}(q)$ is on one fibre and $q+1$ lines of type (ii), and every line has q points.) $\mathrm{AS}(q)$ is a GQ of order $(q-1, q+1)$ [P71], [PT, 3.1.4, 3.2.6]; for q odd it is the Ahrens–Szekeres quadrangle [AS69]. The fibres form a spread.

Fix a non-square $\varepsilon \in \mathbb{F}$ and put $N(b,c) = b^2 - \varepsilon c^2$ (anisotropic: $N(w) = 0$ only for $w = 0$). For $\kappa \in \mathbb{F}^*$ let $\mathcal{E}_\kappa$ be the quadric $X_0X_1 = \kappa N(X_2, X_3)$. It is elliptic (a hyperbolic pair plus an anisotropic binary form), passes through x with tangent plane $x^\perp$, and $\mathcal{E}_\kappa \setminus \{x\} = \{(\kappa N(w) : 1 : w) : w \in \mathbb{F}^2\}$ is a set of q^2 points of $\mathrm{AS}(q)$, one on each fibre.

4.2 Statement

Theorem 4.1 (Theorem A). Let q be odd, ε a non-square and $\kappa_1\kappa_2 = 1/(4\varepsilon)$. Every line of $\mathrm{AS}(q)$ meets $\mathcal{E}_{\kappa_1} \setminus \{x\}$ and $\mathcal{E}_{\kappa_2} \setminus \{x\}$ in exactly two points in total (a point in both counted twice). The two quadrics meet only in x and $e = (0:1:0:0)$. So $\varpi = \mathbf{1}_{\mathcal{E}_{\kappa_1}\setminus x} + \mathbf{1}_{\mathcal{E}_{\kappa_2}\setminus x}$ is a weighted 2-ovoid of $\mathrm{AS}(q)$, with $\varpi(e) = 2$ and $\varpi \in \{0,1\}$ elsewhere.

The choice $\kappa_1 = \frac{1}{2}$, $\kappa_2 = \frac{1}{2\varepsilon}$ gives the two quadrics

$$\mathcal{E}_1 :\ 2X_0X_1 = X_2^2 - \varepsilon X_3^2, \qquad \mathcal{E}_2 :\ 2\varepsilon X_0X_1 = X_2^2 - \varepsilon X_3^2.$$

Corollary 4.2 (Corollary A1). For every odd prime power q, adjoining to $\mathrm{AS}(q)$ one clone of each point of $\mathcal{E}_1 \setminus \{x\}$ and one of each point of $\mathcal{E}_2 \setminus \{x\}$ (so e gets two clones) gives a $(q+2)$-regular incidence structure of diameter 4 with $2q^2(q+2)$ elements. Hence $n_b(q+2, 4) \geq 2q^2(q+2)$.

New vertices. For $P = (\kappa_i N(w) : 1 : w) \in \mathcal{E}_i \setminus \{x\}$ ($i = 1, 2$, $w \in \mathbb{F}^2$), the clone $P^{(i)}$ is adjacent exactly to the $q+2$ lines of $\mathrm{AS}(q)$ through P: the fibre $\{(\kappa_i N(w) + \mu : 1 : w) : \mu \in \mathbb{F}\}$ and, for each $d \in \mathrm{PG}(1,q)$, the line $\ell(\kappa_i N(w), w; d)$. The two clones of e are adjacent to the same $q+2$ lines. Changed old neighbourhoods: the fibre over w_0 gains the clones of $(\kappa_1 N(w_0) : 1 : w_0)$ and $(\kappa_2 N(w_0) : 1 : w_0)$; the line $\ell(a_0, w_0; d)$ gains the clones of its points on $\mathcal{E}_1$ and on $\mathcal{E}_2$, which are given by the roots λ of the quadratic $(\star)$ below with $\kappa = \kappa_1, \kappa_2$ (two clones in total). Old points are unchanged. Every vertex has degree $q+2$.

4.3 Proof

Fibres. The fibre over w_0 meets $\mathcal{E}_\kappa$ only at $\mu = \kappa N(w_0) - a_0$: one point of each quadric.

Lines of type (ii). Let $B = N(w_0 + d) - N(w_0) - N(d) = 2(b_0d_1 - \varepsilon c_0 d_2)$. The point with parameter λ of $\ell(a_0, w_0; d)$ lies on $\mathcal{E}_\kappa$ iff $a_0 + \lambda\rho = \kappa N(w_0 + \lambda d)$, i.e.

$$\kappa N(d)\lambda^2 + (\kappa B - \rho)\lambda + \kappa N(w_0) - a_0 = 0. \qquad (\star)$$

As $N(d) \neq 0$ this is a quadratic, with $1 + \chi(\mathrm{disc}_\kappa)$ roots. A direct expansion gives $B^2 - 4N(w_0)N(d) = 4\varepsilon(b_0d_2 - c_0d_1)^2 = 4\varepsilon\rho^2$, hence

$$\mathrm{disc}_\kappa = (\kappa B - \rho)^2 - 4\kappa N(d)(\kappa N(w_0) - a_0) = 4\kappa N(d)\,(a_0 - a^*_\kappa), \qquad a^*_\kappa = -\frac{4\varepsilon\kappa\rho^2 - 2B\rho + \rho^2/\kappa}{4N(d)}.$$

Then $a^*_{\kappa_1} - a^*_{\kappa_2} = -\frac{\rho^2}{4N(d)}(\kappa_1 - \kappa_2)\big(4\varepsilon - \frac{1}{\kappa_1\kappa_2}\big) = 0$; call the common value a^*. The line meets the two quadrics in

$$\big(1 + \chi(\kappa_1)\chi(N(d)(a_0 - a^*))\big) + \big(1 + \chi(\kappa_2)\chi(N(d)(a_0 - a^*))\big) = 2$$

points, because $\chi(\kappa_1)\chi(\kappa_2) = \chi(1/(4\varepsilon)) = \chi(\varepsilon) = -1$, so $\chi(\kappa_1) + \chi(\kappa_2) = 0$. (If $a_0 = a^*$, the line is tangent to both and meets each in one point.)

Intersection. $\kappa_1 \neq \kappa_2$, since $\kappa_1^2 = 1/(4\varepsilon)$ is impossible for a non-square ε. So $\kappa_1 N(w) = \kappa_2 N(w)$ forces $N(w) = 0$, $w = 0$, and the common points are x and e. □

Proof of Corollary 4.2. Theorem 3.2 with $(s, t) = (q - 1, q + 1)$, $w = \varpi$ of Theorem 4.1 and $\mathcal{M} = \emptyset$: every line has q points and gains exactly two clones, every old point is on $q + 2$ lines, and every clone is on the $q + 2$ lines of its point. The order is $q^3 + q^2(q + 2) + 2q^2 = 2q^2(q + 2)$. □

Remarks. Weights are necessary: an integer program shows that AS(3) has no 1- or 2-ovoid without repeated points and AS(5) has no m-ovoid for $m = 1, \ldots, 4$ (`research/gq4x`). The structure is regular, so no choice of clones and shadows on AS(q) does better at $\Delta = q + 2$. Theorem A beats Delorme's $2q^3 + 3q^2 + 3q + 6$ by $q^2 - 3q - 6 > 0$ for $q \geq 5$.

Cells. $(11, 4)$: 1782 (2013: 1734); $(13, 4)$: 3146 (3064); $(15, 4)$: 5070 (4946). All regular.

Status. Paper proof (above; `NOTE.md` §3, `geometry.tex` Theorem 5.6). Computer-checked: the record graphs 1782, 3146, 5070 (three checkers, `NOTE.md` §7); the normalised form above rebuilt from these coordinates for $q = 3, 5, 7, 9$ (regular, diameter 4). Lean-checked for every finite field of odd order: the model of §4.1 has the GQ counts of order $(q - 1, q + 1)$, two points on at most one line, and the existence half of the GQ axiom, which is all that Theorem B uses (`ASq.isGQst`), the weighted 2-ovoid (`ASq.weighted_two_ovoid`), the bound and its regular form (`ASq.bound`, `ASq.bound_regular`); and the three record graphs (`nb_11_4_v2`, `nb_13_4_v2`, `nb_15_4_v2`).

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5. Proposition C: Singer 2-ovoids of $W(3, q)$, and the $(16, 4)$ graph

5.1 The cyclic model

Let q be odd, $\mathbb{F} = \mathrm{GF}(q) \subset K = \mathrm{GF}(q^2) \subset E = \mathrm{GF}(q^4)$, and g a primitive element of E. E is a 4-dimensional $\mathbb{F}$-space, so $\mathrm{PG}(3, q) = E^*/\mathbb{F}^*$ (any $\mathbb{F}$-basis of E gives homogeneous coordinates); the point $\langle g^i \rangle$ has index i modulo $(q + 1)(q^2 + 1)$. Write $\bar{y} = y^{q^2}$ (the involution of E/K), $k' = k^q$ for $k \in K$, $\mathrm{N}(y) = y\bar{y} \in K$, $\mathrm{Tr} = \mathrm{Tr}_{K/\mathbb{F}}$. Let $d = g^{(q^2+1)/2}$; then $\bar{d} = -d$ and $h = d^2 = g^{q^2+1}$ generates K^*. The form

$$b(y, z) = \mathrm{Tr}_{E/\mathbb{F}}(d\,y\bar{z}) = \mathrm{Tr}\big(d(y\bar{z} - \bar{y}z)\big)$$

is alternating and non-degenerate; $W(3, q)$ is its symplectic quadrangle. Multiplication by g^{q+1} multiplies b by $g^{(q+1)(q^2+1)} \in \mathbb{F}^*$, so it is a collineation of $W(3, q)$. Its point orbits are the classes

$$C_a = \{\langle g^i \rangle : \; i \equiv a \pmod{q + 1}\}, \qquad |C_a| = q^2 + 1,$$

and $\langle y \rangle \in C_a$ iff $\mathrm{N}(y) \in h^a\mathbb{F}^*$ (as $\mathrm{N}(g^i) = h^i$ and $K^*/\mathbb{F}^*$ is cyclic of order $q + 1$, generated by h). These are Ebert's elliptic quadrics partitioning $\mathrm{PG}(3, q)$ [E85].

The K-subspaces yK ($y \in E^*$) are lines of $\mathrm{PG}(3,q)$ (the regular spread), with points $\langle g^{i+(q^2+1)j}\rangle$, $j = 0, \dots, q$. They are totally isotropic: for $k_1, k_2 \in K$, $b(yk_1, yk_2) = \mathrm{Tr}(d\,\mathrm{N}(y)(k_1k_2 - k_1k_2)) = 0$. So $\Sigma = \{yK\}$ is a spread of $W(3,q)$ with q^2+1 lines.

5.2 Statement and proof

Proposition 5.1 (Proposition C). Let q be odd.

- (a) Every line of $\mathrm{PG}(3,q)$ meets each class C_a in at most two points.
- (b) Every totally isotropic line of $W(3,q)$ meets $C_a \cup C_{q-a}$ in exactly two points.

So the $(q+1)/2$ sets $C_a \cup C_{q-a}$, $0 \le a \le (q-1)/2$, are 2-ovoids of $W(3,q)$ partitioning its points (the pairing $a \leftrightarrow q-a$ has no fixed point, as $2a \equiv -1 \pmod{q+1}$ is impossible for $q+1$ even).

Proof. Let $L = \langle y, z\rangle_{\mathbb{F}}$ and $u = sy + tz$ ($s, t \in \mathbb{F}$). Then $\mathrm{N}(u) = A_1 s^2 + \tau st + A_2 t^2$ with $A_1 = \mathrm{N}(y)$, $A_2 = \mathrm{N}(z)$, $\tau = y\bar{z} + \bar{y}z$, all in K. Put $\theta = y\bar{z} - \bar{y}z$. As $\bar{\theta} = -\theta$, $\theta/d \in K$; write $\theta = dk_0$, $k_0 \in K$. Then

$$\mathcal{D} := \tau^2 - 4A_1A_2 = \theta^2 = hk_0^2, \qquad b(y,z) = \mathrm{Tr}(hk_0).$$

Fix $\omega \in K^*$ with $\omega' = -\omega$. For $\mu \in K^*$ put $\nu = \omega/\mu$; the kernel of $k \mapsto \mathrm{Tr}(\nu k)$ is the $\mathbb{F}$-subspace $\mu\mathbb{F}$. So $\langle u\rangle$ lies in $C_\mu := \{\langle u\rangle : \mathrm{N}(u) \in \mu\mathbb{F}^*\}$ iff $(s:t)$ is a zero of the binary form

$$f_\nu(s,t) = \mathrm{Tr}(\nu A_1)s^2 + \mathrm{Tr}(\nu\tau)st + \mathrm{Tr}(\nu A_2)t^2.$$

(a) Suppose $f_\nu \equiv 0$. Then $A_1, \tau, A_2 \in \mu\mathbb{F}$. If $\mathcal{D} \neq 0$, then $hk_0^2 = \mathcal{D} \in \mu^2\mathbb{F}^*$, so $h \in (K^*)^2\mathbb{F}^* = (K^*)^2$ (every element of $\mathbb{F}^*$ is a square in K, since $q-1$ divides $(q^2-1)/2$), but h generates K^*, which has even order. If $\mathcal{D} = 0$, then $\mathrm{N}(sy+tz) = \mu(\alpha_0 s^2 + \beta_0 st + \gamma_0 t^2)$ with $\alpha_0, \beta_0, \gamma_0 \in \mathbb{F}$, $\beta_0^2 = 4\alpha_0\gamma_0$ and $\alpha_0 \neq 0$ (as $A_1 \neq 0$), which vanishes at $(s:t) = (-\beta_0 : 2\alpha_0)$: impossible, since $\mathrm{N}(u) \neq 0$ for $u \neq 0$. So $f_\nu \not\equiv 0$, and it has $1 + \chi(D(\nu)) \le 2$ projective zeros, $D(\nu) = \mathrm{Tr}(\nu\tau)^2 - 4\,\mathrm{Tr}(\nu A_1)\,\mathrm{Tr}(\nu A_2)$.

(b) Expanding with $\mathrm{Tr}(\nu k) = \nu k + \nu' k'$,

$$D(\nu) = \mathrm{Tr}(\nu^2\mathcal{D}) + \nu\nu' P, \qquad P = 2\tau\tau' - 4A_1A_2' - 4A_1'A_2 \in \mathbb{F}.$$

If L is totally isotropic, $\mathrm{Tr}(hk_0) = 0$, so $hk_0 \in \omega\mathbb{F}$ and $\eta := h\mathcal{D} = (hk_0)^2 \in \omega^2\mathbb{F} \subseteq \mathbb{F}$. Then $\mathrm{Tr}(h^2\nu'^2\mathcal{D}) = \eta(h\nu'^2 + h'\nu^2) = hh'\,\mathrm{Tr}(\nu^2\mathcal{D})$, and since $(h\nu')(h\nu')' = hh'\nu\nu'$,

$$D(h\nu') = \mathrm{N}_{K/\mathbb{F}}(h)\,D(\nu).$$

$\mathrm{N}_{K/\mathbb{F}}(h) = h^{q+1}$ generates $\mathbb{F}^*$, so it is a non-square and $\chi(D(h\nu')) = -\chi(D(\nu))$. The class belonging to $h\nu'$ is $\omega/(h\nu') = -\mu'/h \equiv (h\mu)^{-1}$ modulo $\mathbb{F}^*$ (using $\mu^{q+1} \in \mathbb{F}^*$); in indices $a \mapsto -a - 1 \equiv q - a \pmod{q+1}$. Scaling ν by $\mathbb{F}^*$ multiplies D by a square. So L meets $C_a \cup C_{q-a}$ in $(1 + \chi(D(\nu))) + (1 - \chi(D(\nu))) = 2$ points. □

Remark (literature). For q odd, $W(3,q)$ has m-ovoids for every even m and $(q+1)/2$-ovoids [BLP09], and [FWX19] states, citing [BLP09], that $W(3,p^e)$ (p odd) has a partition into 2-ovoids. Cossidente, Culbert, Ebert and Marino [CCEM08] construct m-ovoids of $W(3,q)$ with Singer-type stabilisers. We have not been able to check whether the 2-ovoids $C_a \cup C_{q-a}$ coincide with those; Proposition C is very likely a known object, and we claim only the short self-contained proof. The proof uses only that d^2 is a non-square in K: for any δ_0 with $\bar{\delta}_0 = -\delta_0$, $C_a \cup C_b$ is a 2-ovoid of the form $\mathrm{Tr}_{E/\mathbb{F}}(\delta_0 y\bar{z})$ when $h^{a+b}\delta_0^2 \in \mathbb{F}^*$.

5.3 Corollary C1, written in $W(3,q)$

Corollary 5.2 (Corollary C1). Let q be odd, $0 \le a \le (q-1)/2$, $\mathcal{O} = C_a \cup C_{q-a}$ and Σ the regular spread $\{yK\}$. Adjoin to $W(3,q)$

- one clone P' of each point $P \in \mathcal{O}$, incident with the $q+1$ totally isotropic lines on P (the lines $\langle P, Z\rangle$ with $b(P,Z) = 0$);

- two shadow lines L', L'' for each $L \in \Sigma$, each incident with the $q+1$ points of L.

The result is bipartite, has diameter 4, maximum degree $q+3$ (old points and old lines have degree $q+3$, clones and shadow lines $q+1$) and $2(q^2+1)(q+3)$ elements. Joining clone c_i to shadow lines y_{2i}, y_{2i+1} (indices modulo $2(q^2+1)$, any numbering) makes it $(q+3)$-regular with diameter 4. Hence $n_b(q+3,4) \geq 2(q^2+1)(q+3)$, with a regular graph.

Proof. $W(3,q)$ is a GQ of order (q,q). By Proposition 5.1(b), $w = \mathbf{1}_{\mathcal{O}}$ is a 2-ovoid ($= \Delta - s - 1$ for $\Delta = q+3$), and $\mathcal{M} = 2\Sigma$ is a 2-cover (every point is on exactly one spread line). Apply Corollary 3.3 with $k = 2$. □

This is the point–line dual of the construction in `NOTE.md` §5.4 (clones of an ovoid of $Q(4,q)$ and the dual 2-cover), so it needs no Klein correspondence. (The $(16,4)$ record file was built in the $Q(4,13)$ model; it has the same order and degrees.) More generally, j copies of the spread and $j/2$ of the 2-ovoid give $n_b(q+1+j,4) \geq 2(q^2+1)(q+1+j)$ for even j.

Corollary 5.3 (Corollary B2: all degrees). For every prime power q and $\Delta \geq q+1$ there is a Δ-regular incidence structure of diameter 4 with $2\Delta(q^2+1)$ elements if q is even or $\Delta - q - 1$ is even, and $2\Delta q^2$ elements otherwise. In particular $n_b(\Delta,4) \geq 2q^2\Delta$ for every prime power $q \leq \Delta - 1$.

Proof. Corollary 3.3 with $j = \Delta - q - 1$. q even: $W(3,q)$ has an ovoid $\mathcal{O}_0$ and a spread Σ [PT, 3.4.1]; take $w = j\mathbf{1}_{\mathcal{O}_0}$, $\mathcal{M} = j\Sigma$. q odd, j even: $w = (j/2)\mathbf{1}_{\mathcal{O}}$ with $\mathcal{O}$ a 2-ovoid of Proposition 5.1, $\mathcal{M} = j\Sigma$. q odd, j odd: then $\Delta \geq q+2$; in AS(q) take $(\Delta - q)/2$ copies of the weighted 2-ovoid of Theorem 4.1 and $\Delta - q - 2$ copies of the fibre spread. □

Cells. $(16,4)$: $q = 13$, $5440 = 2 \cdot 170 \cdot 16$ (2013: 5134), regular. In the range $\Delta \leq 16$ this is the only open cell with $\Delta - 3$ an odd prime power. Without stand-ins, 5440 is the most a copy graph on $W(3,13)$ or $Q(4,13)$ of maximum degree 16 can have (each line takes at most 2 clones and each point at most 2 shadow lines).

Status. Paper proof (above; `NOTE.md` §5, `geometry.tex` Proposition 5.9). Computer-checked: exactly the pairs $\{a, q-a\}$ are 2-ovoids for $q = 3, 5, 7, 11, 13$ (`research/gq4y/w3_singer.py`) and $q = 9, 25, 27$ (`w3_singer_pp.py`); the 5440 graph (three checkers); the $W(3,q)$ form of Corollary 5.2 for $q = 3, 5$. Lean-checked, in exactly this cyclic model and directly in $W(3,q)$ (no Klein correspondence): for every field E of order q^4, q odd, with a chosen primitive element g, Proposition C(b) (`SingerC.Setting.propC_subspace`, `SingerC.propC`), the spread $\{yK\}$ (`SingerC.W3.spread_exists_unique`), Corollary C1 for every even $j \geq 2$ (`SingerC.corC1`) and its regular form (`SingerC.Setting.regular_W3`); such data exist for every odd prime power, but the library constructs them explicitly only for $q = 13$, where it gives the regular 5440 graph (`SingerC.regular5440`). The record file is also kernel-checked (`nb_16_4_v2`). See Section 9.

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6. Theorem D: the split Cayley hexagon, Hermitian spreads and whiskers (diameter 6)

Throughout this section q is odd (q even: Remark 6.10).

6.1 Coordinates

$Q(6,q)$ is the quadric $X_0X_4 + X_1X_5 + X_2X_6 = X_3^2$ of PG$(6,q)$, with polar form B. The split Cayley hexagon $H(q)$ has as points all points of $Q(6,q)$, and as lines the lines of $Q(6,q)$ whose Grassmann coordinates $p_{ij} = x_iy_j - x_jy_i$ satisfy Tits' equations

$$p_{12} = p_{34}, \quad p_{54} = p_{32}, \quad p_{20} = p_{35}, \quad p_{65} = p_{30}, \quad p_{01} = p_{36}, \quad p_{46} = p_{31}.$$

It is a generalized hexagon of order (q,q) [T59], [VM98, §2.4]: $(q+1)(q^4+q^2+1)$ points and as many lines, any two elements at distance ≤ 6, girth 12. The lines on a point y are the $q+1$ lines on y in a totally singular plane π_y. Let $v = \langle e_1, e_6 \rangle$ (a line of $H(q)$: no Tits equation involves p_{16}). Fix $c \in \mathbb{F}$ with $-c$ a non-square. For $z = (\alpha, \beta) \in \mathbb{F}^2$ let

$$\Pi_z : \; X_0 + cX_4 + \alpha X_2 + \beta X_5 = 0, \qquad \Pi = \Pi_{(0,0)},$$

and let S_z be the set of lines of $H(q)$ contained in Π_z; $S = S_{(0,0)}$. Every Π_z contains v.

Lemma 6.1. For every $z \in \mathbb{F}^2$:

- (a) $\Pi_z \cap Q(6,q)$ is an elliptic quadric $Q^-(5,q)$.
- (b) Every point of $H(q)$ in Π_z is on exactly one line of S_z. So $|S_z| = q^3+1$, the lines of S_z are pairwise opposite, and $v \in S_z$ (S_z is a Hermitian spread [T80]).
- (c) A line of $H(q)$ not in S_z meets Π_z in exactly one point.
- (d) The points of $H(q)$ at distance ≤ 3 from v are exactly the points of $Q(6,q)$ with $X_2 = X_5 = 0$; among them only the points of v lie in Π_z.
- (e) $S_z \cap S_{z'} = \{v\}$ for $z \neq z'$.

Proof. (a) On Π_z substitute $X_0 = -cX_4 - \alpha X_2 - \beta X_5$: $X_0X_4 + X_1X_5 + X_2X_6 - X_3^2 = (X_1 - \beta X_4)X_5 + X_2(X_6 - \alpha X_4) - (X_3^2 + cX_4^2)$, two hyperbolic planes plus the anisotropic form $X_3^2 + cX_4^2$. So the form on Π_z is non-degenerate of Witt index 2.

(b) Let $y \in \Pi_z \cap Q(6,q)$. By (a) $\Pi_z \cap Q(6,q)$ contains no totally singular plane, so $\pi_y \not\subseteq \Pi_z$ and $\pi_y \cap \Pi_z$ is one line through y: the only line of $H(q)$ on y inside Π_z. $Q^-(5,q)$ has $(q+1)(q^3+1)$ points, so $|S_z| = q^3+1$. Two lines of S_z are not concurrent (uniqueness). If two were at distance 4, a line N of $H(q)$ would meet both in distinct points, so $N \subseteq \Pi_z$, $N \in S_z$, concurrent with them: a contradiction. So they are opposite.

(c) A line not in the hyperplane Π_z meets it in one point.

(d) Let $y = ae_1 + be_6$ be on v and yu a line of $H(q)$. Then $p_{12} = p_{34}$ gives $au_2 = 0$, $p_{65} = p_{30}$ gives $bu_5 = 0$, and $B(y,u) = 0$ gives $au_5 + bu_2 = 0$; so $u_2 = u_5 = 0$. Conversely $Q(6,q) \cap \{X_2 = X_5 = 0\}$ is $X_0X_4 = X_3^2$, a cone with vertex v over a conic, with $(q+1) + (q+1)q^2$ points: exactly the number of points at distance ≤ 3 from v ($q+1$ on v, and q more on each of the $q(q+1)$ other lines meeting v). If such a point is in Π_z, then $X_0 = -cX_4$ and $X_3^2 + cX_4^2 = 0$, so $X_0 = X_3 = X_4 = 0$ and the point is on v.

(e) If $\ell \in S_z \cap S_{z'}$, $\ell \neq v$, then $(\alpha - \alpha')X_2 + (\beta - \beta')X_5 = 0$ on ℓ, so some point of ℓ has $X_2 = X_5 = 0$ and is at distance ≤ 3 from v by (d); but ℓ and v are opposite by (b), so every point of ℓ is at distance 5 from v. □

6.2 Delorme's levels and whiskers

Let $\mathcal{P}(v)$ be the points of v, $\mathcal{L}_v$ the q^2+q lines $\neq v$ meeting v, and $\mathcal{P}_\Pi$ the $(q+1)(q^3+1)$ points of $H(q)$ in Π. A point is *free* if it is not in Π and is at distance 5 from v; by Lemma 6.1(d) there are $(q+1)(q^4+q^2+1) - (q+1) - q^2(q+1) - q^3(q+1) = q^3(q^2-1)$ of them.

Definition 6.2. A *whisker set* is a set W of lines of $H(q)$, each opposite v and not in S, such that no free point is on two lines of W. It is *exact* if every free point is on exactly one line of W.

A line opposite v and not in S meets Π in one point, not on v (Lemma 6.1(c),(d)), and its other q points are free (every point of a line opposite v is at distance 5 from v). So $|W| \le q^3(q^2-1)/q = q^2(q^2-1)$, with equality iff W is exact.

Theorem 6.3. Let W be a whisker set and $m \ge 2$. Put

$$\mathcal{A}_2 = \{v\}, \qquad \mathcal{A}_1 = \{v\} \cup \mathcal{P}(v) \cup S, \qquad \mathcal{A}_0 = \mathcal{A}_1 \cup \mathcal{P}_\Pi \cup \mathcal{L}_v \cup W.$$

Then $\mathcal{A}$ satisfies (L0), (L1) and (C_6). Hence $H(q)^{(m)}_{\mathcal{A}}$ is bipartite with maximum degree $q+m$, diameter 6, and

$$2(q+1)(q^4+q^2+1) + (m-1)\big(q^4+2q^3+q^2+2q+2+|W| + m(q^3+q+2) + m^2\big)$$

elements.

Proof. (L1), $i=1$: the elements on v are its points ($\in \mathcal{A}_1$); a point of v is on one element of $\mathcal{A}_2$, a line of $S \setminus \{v\}$ on none. (L1), $i=0$: the elements incident with $\mathcal{A}_1$ are the points of v, the lines of $\mathcal{L}_v$ and the points of Π, all in $\mathcal{A}_0$; a line of $\mathcal{L}_v$ meets v in one point and contains no other point of $\mathcal{A}_1$; a point of $\mathcal{P}_\Pi \setminus \mathcal{P}(v)$ is on exactly

one line of S (Lemma 6.1(b)) and not on v; a whisker has no point on v. (L0): a point outside $\mathcal{A}_0$ is off Π. If it is at distance 3 from v, it is on exactly one line of $\mathcal{L}_v$ (two would close a circuit of length ≤ 6) and on no whisker (a whisker through it would be at distance ≤ 4 from v). If it is free, it is on no line of $\mathcal{L}_v$ and on at most one whisker. A line outside $\mathcal{A}_0$ is not in Π (the lines in Π are those of S), so it meets Π in one point, its only point in $\mathcal{A}_0$. (C_6): a line of $\mathcal{L}_v$ has no point of Π other than its point on v (Lemma 6.1(d)), and a whisker has exactly one point of Π, not on v. So the components of the substructure on $\mathcal{A}_0$ are

$$\{v\} \cup \mathcal{P}(v) \cup \mathcal{L}_v \qquad \text{and} \qquad \{n\} \cup \mathcal{P}(n) \cup \{w \in W : w \text{ meets } n\} \quad (n \in S \setminus \{v\}),$$

trees of diameter 4 (the ball of radius 2 about v; a line, its points and the whiskers hanging from them). Lemma 1.3 gives (C_6), and Theorem 1.2 gives degree $\le q+1+(m-1)$, diameter ≤ 6 and the count: $|\mathcal{A}_2| = 1$, $|\mathcal{A}_1| = (q^3+1)+(q+1)$, $|\mathcal{A}_0| = (q^3+1)+(q^2+q)+|W|+(q+1)(q^3+1)$. The diameter is exactly 6: a free point p (level -1) has q^5 opposite points, and only $(q+1)(q^3+1) < q^5$ points lie in $\mathcal{A}_0$, so some opposite point p' has level -1 and $d(p,p') \ge 6$ by projection. □

New vertices: the neighbourhood of every version. Write x^w for the versions ($a, b, c \in [m]$). Every element of $H(q)$ appears once in this table.

element (level)	versions	neighbours of the version	degree
v (2)	v^{abc}	p^{ab} for the $q+1$ points p of v	$q+1$
point p of v (1)	p^{ab}	$v^{ab1}, \dots, v^{abm}$; L^a for the q lines $L \in \mathcal{L}_v$ on p	$q+m$
line $n \in S \setminus \{v\}$ (1)	n^{ab}	p^a for the $q+1$ points p of n	$q+1$
point $p \in \mathcal{P}_\Pi \setminus \mathcal{P}(v)$ (0)	p^a	$n^{a1}, \dots, n^{am}$ for the line $n \in S$ on p; w^a for each whisker w on p; every other line on p	$q+m$
line $L \in \mathcal{L}_v$ (0)	L^a	$p^{a1}, \dots, p^{am}$ for $p = L \cap v$; the q other points of L	$q+m$
whisker $w \in W$ (0)	w^a	p^a for $p = w \cap \Pi$; the q free points of w	$q+1$
point $r \notin \Pi$ (-1)	r	all m versions of the line of $\mathcal{L}_v \cup W$ on r (if any); every other line on r	$q+m$, or $q+1$ if r is free and on no whisker
line $N \notin \mathcal{A}_0$ (-1)	N	all m versions of its point in Π (not on v); its q other points	$q+m$

An element of level k has m^{k+1} versions. For $m = 2$ ($\Delta = q+2$) the number of new elements is $|\mathcal{A}_0|+2|\mathcal{A}_1|+4|\mathcal{A}_2|$ (Theorem 1.2(c)).

6.3 An exact whisker set for every odd q

For a point X of $H(q)$ in Π, not on v, put

$$\nu(X) = \frac{X_2X_3 - X_4X_5}{cX_2^2 + X_5^2}.$$

The denominator vanishes only if $X_2 = X_5 = 0$, i.e. (Lemma 6.1(d)) only on v. Let

$$G(a,b) = (c_0a + c_1b)(a^2 + cb^2) - b^3, \qquad F(z) = \frac{G(\alpha,\beta)}{\alpha^2 + c\beta^2} \quad (z = (\alpha,\beta) \ne 0),$$

with $c_0, c_1 \in \mathbb{F}$ chosen so that G has no zero on $\mathbb{F}^2 \setminus \{0\}$.

Lemma 6.4. Such c_0, c_1 exist for every odd q.

Proof. $G(a,0) = c_0 a^3$, so $c_0 \neq 0$ is needed. For $b \neq 0$ and $t = a/b$, $G(a,b) = b^3\big((c_0 t + c_1)(t^2 + c) - 1\big)$, and $t^2 + c \neq 0$. So G has a zero with $b \neq 0$ iff (c_0, c_1) lies on one of the q lines $c_0 t + c_1 = 1/(t^2 + c)$, $t \in \mathbb{F}$, of the (c_0, c_1)-plane; each has $q - 1$ points with $c_0 \neq 0$. The lines for $t = 0$ and $t = 1$ meet at $c_0 = 1/(1 + c) - 1/c = -1/(c(1 + c)) \neq 0$ ($c \neq -1$ as $-c$ is a non-square). So these lines cover at most $q(q-1) - 1$ of the $q(q-1)$ points with $c_0 \neq 0$. □

One valid choice for each q in the table ($\mathbb{F}_9 = \mathbb{F}_3[i]$, $i^2 = -1$):

q	3	5	7	9	11	13
c	1	2	1	$1 + i$	1	2
(c_0, c_1)	(1,2)	(1,0)	(1,0)	$(i, 2i)$	(1,3)	(1,1)

Theorem 6.5 (exact whiskers). Let q be odd, $-c$ a non-square, c_0, c_1 as in Lemma 6.4. Then

$$W = \big\{\ell \in S_z \setminus \{v\} :\ z \in \mathbb{F}^2 \setminus \{0\},\ \nu(\ell \cap \Pi) = F(z)\big\}$$

is an exact whisker set: every free point lies on exactly one line of W, and $|W| = q^2(q^2 - 1)$.

(Here $\ell \cap \Pi$ is one point not on v: $\ell \subseteq \Pi_z$, $\ell \neq v$, so $\ell \notin S$ by Lemma 6.1(e) and ℓ is opposite v by Lemma 6.1(b).) The proof uses a complex structure on each Π_z. Write a point of Π_z as $x = (x_1, \ldots, x_6)$, meaning $(X_0, \ldots, X_6) = (-cx_4 - \alpha x_2 - \beta x_5, x_1, \ldots, x_6)$; on Π_z the quadratic form is $Q_z(x) = x_1x_5 + x_2x_6 - x_3^2 - cx_4^2 - \alpha x_2x_4 - \beta x_4x_5$, with polar form B_z. Put $\mathrm{num}(y) = y_2y_3 - y_4y_5$ and $\mathrm{den}(y) = cy_2^2 + y_5^2$, and define the linear map

$$J_z x = \big(-\beta x_3 + \alpha x_4 - x_6,\ x_5,\ -cx_4 - \beta x_5,\ x_3 + \beta x_2,\ -cx_2,\ cx_1 + \alpha\beta x_2 + \alpha x_3 + c\beta x_4 + \beta^2 x_5\big);$$

for $z = 0$, $J_0 x = (-x_6, x_5, -cx_4, x_3, -cx_2, cx_1)$.

Lemma 6.6 (complex structure). For every z:

- (a) $J_z^2 = -c$, $Q_z(J_z x) = cQ_z(x)$ and $B_z(x, J_z x) = 0$.
- (b) The six differences $p_{12} - p_{34}$, $p_{54} - p_{32}$, $p_{20} - p_{35}$, $p_{65} - p_{30}$, $p_{01} - p_{36}$, $p_{46} - p_{31}$ of the Grassmann coordinates of $x \wedge J_z x$ equal $Q_z(x)$ times $1, 0, 0, -c, \alpha, -\beta$.
- (c) For every point x of $Q(6,q) \cap \Pi_z$, the line $\langle x, J_z x\rangle$ is the line of S_z through x.
- (d) $\mathrm{num}(sx + tJ_0 x) = (s^2 + ct^2)\,\mathrm{num}(x)$ and $\mathrm{den}(sx + tJ_0 x) = (s^2 + ct^2)\,\mathrm{den}(x)$. So ν is constant on every line of $S \setminus \{v\}$.

Proof. (a), (b) and (d) are polynomial identities with integer coefficients, checked by direct expansion (for instance, for $z = 0$: $Q_0(J_0x) = cx_2x_6 + cx_1x_5 - c^2x_4^2 - cx_3^2 = cQ_0(x)$, and $p_{12} = x_1x_5 + x_2x_6$, $p_{34} = x_3^2 + cx_4^2$, so $p_{12} - p_{34} = Q_0(x)$; in (d), $\mathrm{num}(J_0x) = c\,\mathrm{num}(x)$, $\mathrm{den}(J_0x) = c\,\mathrm{den}(x)$, and the polar forms $x_2y_3 + x_3y_2 - x_4y_5 - x_5y_4$ and $2cx_2y_2 + 2x_5y_5$ vanish at $y = J_0x$). We re-checked all of them with sympy for this note (`polish/jz.py`). (c) J_z has no eigenvector ($-c$ is a non-square), so $x, J_z x$ span a line; if $Q_z(x) = 0$, then by (a) the whole line is singular and by (b) its Grassmann coordinates satisfy Tits' equations, so it is a line of $H(q)$ in Π_z through x, the one of Lemma 6.1(b). (d) By (c), a line $n \in S \setminus \{v\}$ is $\{sx + tJ_0x\}$ for any of its points x, and $s^2 + ct^2 \neq 0$. □

Let p be a free point. By Lemma 6.1(d), $(p_2, p_5) \neq (0,0)$; as $p \notin \Pi$ we normalise $p_0 + cp_4 = 1$. Then $p \in \Pi_z$ iff z lies on the affine line

$$A_p = \{(\alpha, \beta) :\ \alpha p_2 + \beta p_5 = -1\} \subseteq \mathbb{F}^2 \setminus \{0\},$$

which has q points and direction $(p_5, -p_2)$. For $z \in A_p$ let $\ell_p(z)$ be the line of S_z on p and $X(z) = \ell_p(z) \cap \Pi$.

Lemma 6.7 (key identity). For $z = (\alpha, \beta) \in A_p$,

$$\Phi_p(z) := \nu(X(z)) + \frac{\beta^3}{\alpha^2 + c\beta^2} = \frac{p_2p_3 - p_4p_5 + \beta p_2^2}{cp_2^2 + p_5^2}.$$

So Φ_p is affine on A_p, with increment $-p_2^3/(cp_2^2 + p_5^2)$ along $(p_5, -p_2)$.

Proof. Put $\Theta = cp_2^2 + p_5^2 \neq 0$ and $\gamma = -\alpha p_5 + c\beta p_2$. On Π_z the function $\lambda(y) = y_0 + cy_4$ equals $-\alpha y_2 - \beta y_5$, and $\Pi = \{\lambda = 0\}$. We have $\lambda(p) = 1$ and $\lambda(J_z p) = -\alpha p_5 + c\beta p_2 = \gamma$. By Lemma 6.6(c), $\ell_p(z) = \langle p, J_z p\rangle$, so $X(z) = \gamma p - J_z p$, with

$$X_2 = \gamma p_2 - p_5, \quad X_3 = \gamma p_3 + cp_4 + \beta p_5, \quad X_4 = \gamma p_4 - p_3 - \beta p_2, \quad X_5 = \gamma p_5 + cp_2.$$

Expanding, $\mathrm{den}(X) = (\gamma^2 + c)\Theta$ and $\mathrm{num}(X) = (\gamma^2 + c)(p_2p_3 - p_4p_5) + \beta(2\gamma p_2p_5 + cp_2^2 - p_5^2)$. Solving $\alpha p_2 + \beta p_5 = -1$, $-\alpha p_5 + c\beta p_2 = \gamma$ (determinant Θ) gives $\alpha = -(cp_2 + \gamma p_5)/\Theta$, $\beta = (\gamma p_2 - p_5)/\Theta$, hence $\alpha^2 + c\beta^2 = (\gamma^2 + c)/\Theta$ ($\neq 0$) and $\beta^3/(\alpha^2 + c\beta^2) = \beta(\gamma p_2 - p_5)^2/((\gamma^2 + c)\Theta)$. The two β-terms add up to $\beta\big(2\gamma p_2p_5 + cp_2^2 - p_5^2 + (\gamma p_2 - p_5)^2\big)/((\gamma^2 + c)\Theta) = \beta p_2^2/\Theta$. Along $(p_5, -p_2)$, β changes by $-p_2$. □

Proof of Theorem 6.5. Let p be free. The lines of $\bigcup_{z\neq 0} S_z \setminus \{v\}$ on p are exactly the q lines $\ell_p(z)$, $z \in A_p$, and they are distinct (a common line would lie in $S_z \cap S_{z'} = \{v\}$); since $S_z \cap S_{z'} = \{v\}$, each line of W belongs to a single S_z. As $F(z) = c_0\alpha + c_1\beta - \beta^3/(\alpha^2 + c\beta^2)$, we have $\ell_p(z) \in W$ iff $\Phi_p(z) - (c_0\alpha + c_1\beta) = 0$. The left side is affine on A_p with increment

$$-\frac{p_2^3}{\Theta} - (c_0p_5 - c_1p_2) = -\frac{G(p_5, -p_2)}{\Theta} \neq 0$$

along $(p_5, -p_2)$, so it vanishes at exactly one of the q points of A_p: p is on exactly one line of W. The lines of W are opposite v and not in S. Counting incidences between free points and W gives $|W| = q^3(q^2 - 1)/q$. □

6.4 Theorem D

Corollary 6.8 (Theorem D). For every odd prime power q and every $m \geq 2$, the structure of Theorem 6.3 with the exact whisker set of Theorem 6.5 is bipartite, has maximum degree $q + m$ and diameter 6, and

$$n_b(q + m, 6) \geq 2(q + 1)(q^4 + q^2 + 1) + (m - 1)\big(2q^4 + 2q^3 + 2q + 2 + m(q^3 + q + 2) + m^2\big).$$

In particular $n_b(q + 2, 6) \geq 2(q + 1)(q^4 + q^2 + 1) + 2q^4 + 4q^3 + 4q + 10$ and $n_b(q + 3, 6) \geq 2(q + 1)(q^4 + q^2 + 1) + 4q^4 + 10q^3 + 10q + 34$.

With W exact, every element outside $\mathcal{A}_0$ is incident with exactly one element of $\mathcal{A}_0$; by the counting bound for doublings of a generalized n-gon of order (q, q) with N points, $q|\mathcal{A}_0| + 2k \leq 2N$ (k the number of components of $\mathcal{A}_0$; `geometry.tex` Prop. 3.6), the single-level part is as large as possible (here $k = q^3 + 1$).

Remark 6.9 (Delorme's construction: the 2013 values). $W = \emptyset$ is Delorme's doubling of $H(q)$ along a Hermitian spread [D85a, D85b]. At $m = 2$ it has $2(q + 1)(q^4 + q^2 + 1) + q^4 + 4q^3 + q^2 + 4q + 10$ elements: 8992, 142464, 374452, 842048 at $q = 5, 9, 11, 13$; at $q = 13$, $m = 3$, it gives 884062. These are the 2013 table values at $\Delta = 7, 11, 13, 15, 16$. The exact whiskers add $(m - 1)q^2(q^2 - 1)$.

Remark 6.10 (even q). In characteristic 2 the quadric $X_0X_4 + X_1X_5 + X_2X_6 + X_3^2 = 0$ has the nucleus e_3, and the hyperplanes above contain it. `geometry.tex` §7 uses instead $\Pi_z : X_3 = X_0 + \delta X_4 + \alpha X_2 + \beta X_5$ with $\mathrm{Tr}(\delta) = 1$, $\nu(X) = (X_2X_3 + X_2X_4 + X_4X_5)/(\delta X_2^2 + X_2X_5 + X_5^2)$, $G(a, b) = (c_0a + c_1b)(a^2 + ab + \delta b^2) + b^3$, and $J_z^2 + J_z + \delta = 0$; good (c_0, c_1) exist iff $q \geq 4$. The proof is parallel and is written out there (Lemmas 7.1, 7.3, 7.4 and Theorem 7.5); we do not repeat it. For $q = 2$ an exact whisker set of 12 lines was found by computer. Even q improves no cell with $\Delta \leq 16$ ($q + 1$ is then a prime power, and $q + 3$ is better served by the odd $q + 1$).

Worked counts. $q = 5$, $m = 2$: $2 \cdot 6 \cdot 651 = 7812$ elements of $H(5)$, $|\mathcal{A}_0| = 1512$ (of which $|W| = 600$), $|\mathcal{A}_1| = 132$, $|\mathcal{A}_2| = 1$, so $7812 + 1512 + 2 \cdot 132 + 4 = 9592$. $q = 13$, $m = 3$: 804468 elements of $H(13)$, $|\mathcal{A}_0| = 61544$ ($|W| = 28392$), $|\mathcal{A}_1| = 2212$, so $804468 + 2(61544 + 3 \cdot 2212 + 9) = 940846$.

Cells. $m = 2$: $(7, 6)$ 9592 (2013: 8992), $(11, 6)$ 148944 (142464), $(13, 6)$ 388972 (374452), $(15, 6)$ 870440 (842048). $m = 3$, $q = 13$: $(16, 6)$ 940846 (884062). These graphs are not regular (Section 7).

Status. Paper proof (above; `geometry.tex` §6). Computer-checked: the graphs for $q = 5, 7, 9, 11, 13$ (and $q = 13$, $m = 3$) were built and their exact diameter computed by all-pairs BFS; exactness of W checked on all free points for $q = 3, 5, 7$; the identities of Lemmas 6.6–6.7 re-checked symbolically. Lean-checked for every finite field of odd characteristic: $H(q)$ in these coordinates is a generalized hexagon in the graph sense (`Hexagon.genPolygonGraph_hexagon`), Theorem 6.5 (`HexagonD.theorem4`), Theorem 6.3 for any whisker set (`HexagonD.whiskers_bound_hexagon`), Corollary 6.8 (`HexagonD.theoremD`, `theoremD_two`), Lemma 6.4 (`exists_goodCubic`), Lemma 6.7 (`HexagonD.Identity.key_identity`) and Lemma 6.6(d) (`nu_eq_of_mem_sline`); for even $q \geq 4$ (`HexagonEven.theorem4_even`, `theoremD_even`) and $q = 2$, $m = 2$ (`HexagonEven.theoremD_q2`); and the 9592 graph itself (`nb_7_6`). See Section 9.

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7. Theorem R: regular graphs of diameter 6

The structure of Corollary 6.8 has equally many points and lines in $\mathcal{A}_0$ (exact whiskers force this), but $\mathcal{A}_1$ has $q^3 + 1$ lines and only $q + 1$ points, and $\mathcal{A}_2$ is a line. So it has $(m-1)m(q^3 - q + m)$ more lines than points. We delete repeated lines and then add anti-flags.

Lemma 7.1 (repeated lines). In $H(q)^{(m)}_{\mathcal{A}}$, for $n \in S \setminus \{v\}$ and $a \in [m]$ the m lines $n^{a1}, \ldots, n^{am}$ all have the point set $\{p^a : p \in n\}$; for $a, b \in [m]$ the lines $v^{ab1}, \ldots, v^{abm}$ all have the point set $\{p^{ab} : p \in v\}$ (table of §6.2). So there are $mq^3 + m^2$ classes of m repeated lines.

Lemma 7.2 (completion). Let a finite incidence structure have maximum degree $\leq \Delta$, every element of degree $\geq \Delta - r$, and $\sum_{\text{points}}(\Delta - \deg) = \sum_{\text{lines}}(\Delta - \deg) =: T$. If $T > 2r\Delta$, there is a set J of anti-flags such that adding J makes it Δ-regular.

Proof. Choose a multiset J of point–line pairs in which each element x occurs $\Delta - \deg x$ times (possible as the sums agree). Call an entry bad if it is a flag or a repeat of another entry, and take J with fewest bad entries. If (p, L) is bad: L is joined (in $\mathrm{I} \cup J$, with multiplicity) to Δ points, each in $\leq r$ entries of J; likewise p. So at most $2r\Delta < T$ entries (p', L') have p' joined to L or L' joined to p, and some entry (p', L') has neither (it is not (p, L) itself). Replacing $(p, L), (p', L')$ by $(p, L'), (p', L)$ removes a bad entry and creates none. So the minimum is 0. □

Theorem 7.3 (Theorem R). For every odd prime power q and $m \geq 2$ there is a $(q + m)$-regular incidence structure of diameter at most 6 with

$$2(q+1)\big(q^4 + q^2 + 1 + (m-1)(q^3 + m + 1)\big)$$

elements: $2(q + 1)(q^4 + q^3 + q^2 + 4)$ for $m = 2$ and $2(q + 1)(q^4 + 2q^3 + q^2 + 9)$ for $m = 3$. For the five table instances the order exceeds $M_b(q + m, 5)$, so the diameter is exactly 6.

Proof. Start from Corollary 6.8. Fix a set R of q lines of $S \setminus \{v\}$, and delete the lines v^{abc} with $c \geq 2$ ($(m-1)m^2$ lines) and n^{ab} with $b \geq 2$ for $n \notin R$ ($(m-1)m(q^3 - q)$ lines): in total $(m-1)m(q^3 - q + m)$, the excess, so points and lines are now equal in number. Each class of Lemma 7.1 keeps at least one line, so by Lemma 0.2(b) no distance between remaining elements increases. After the deletion the degrees are: p^{ab} $(p \in v)$ $q + 1$; p^a on a line $n \notin R$ of S: $q + 1$; p^a on $n \in R$: $q + m$; v^{ab1}, n^{a1} $(n \notin R)$, n^{ab} $(n \in R)$ and w^a $(w \in W)$: $q + 1$; all others as in §6.2 $(q + m)$. So every deficit is at most $r = m - 1$, and T is at least the deficit of the $m|W|$ whisker versions: $T \geq (m-1)mq^2(q^2 - 1) > 2(m-1)(q+m)$. Lemma 7.2 completes the structure; added flags join a point to a line, so the graph stays bipartite, and they do not increase distances. The number of points is $(q+1)(q^4 + q^2 + 1) + (m-1)\big((q+1)(q^3+1) + m(q+1)\big)$. □

Neighbourhoods. Every neighbourhood is that of the table in §6.2, minus the deleted lines listed above, plus the $(m - 1)$ (for each deficient element, exactly its deficit) completing anti-flags. The completing anti-flags are not given by a formula: Lemma 7.2 proves they exist, and `research/p4/regular/regularize.py` finds them by a seeded swap procedure (it uses its own deletion choice, not necessarily R above). Neither the maximum degree nor the diameter depends on that choice.

Cells (regular graphs). $(7,6)$ 9348, $(11,6)$ 147500, $(13,6)$ 386328, $(15,6)$ 866068, $(16,6)$ 927724 ($q = 13$, $m = 3$). Each beats the 2013 value of its cell (8992, 142464, 374452, 842048, 884062), and Delorme's graphs behind those values are not regular (their point side is larger).

Status. Paper proof (above; `geometry.tex` Theorem 6.13; `research/p4/regular/REPORT.md`). Computer-checked: all five regular graphs are exactly regular with equal sides and diameter exactly 6 (independent all-pairs checks). Lean-checked: `HexagonD.theoremR`, `theoremR_two` (every finite field of odd characteristic); `HexagonEven.theoremR`, `theoremR_two` (even $q \geq 4$; their docstrings wrongly say "odd q", the statements are for characteristic 2); `HexagonEven.theoremR_q2` ($q = 2$, $m = 2$, 192 vertices). See Section 9.

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8. The Haar lemma (diameter 5)

This one is group-theoretic, not geometric; it is included because it gives the $(7,5)$ and $(9,5)$ cells.

Definition 8.1. For a finite group G and $S \subseteq G$, the *Haar graph* $H(G,S)$ has vertex set $G \times \{A,B\}$ and edges $(g,A) \sim (sg,B)$ for $g \in G$, $s \in S$. So

$$N(g,A) = \{(sg,B) : s \in S\}, \qquad N(h,B) = \{(s^{-1}h,A) : s \in S\},$$

it is bipartite and $|S|$-regular on $2|G|$ vertices, and right multiplication by G acts on it as automorphisms.

Lemma 8.2 (Haar lemma). If $S \neq \emptyset$ and every element of G is a product $a^{-1}bc^{-1}d$ and also a product $ab^{-1}cd^{-1}$ with $a,b,c,d \in S$, then $H(G,S)$ has diameter at most 5.

Proof. Two steps from (g,A) reach (tg,B) and then $(s^{-1}tg,A)$, $s,t \in S$: they multiply on the left by $s^{-1}t$. So the A-vertices within distance 4 of (g,A) include $\{(a^{-1}bc^{-1}d\,g,A)\} = G \times \{A\}$. Likewise two steps from (h,B) reach $(s^{-1}h,A)$ and $(ts^{-1}h,B)$, so the B-vertices within distance 4 of (h,B) are all of $G \times \{B\}$. For (g,A) and (h,B): one step to (sg,B) ($s \in S$), then at most 4. □

More generally (Lean `haar_ediam_le_odd`, `haar_ediam_le_even`): if every element is a product of at most n factors $s^{-1}t$ and of at most n factors ab^{-1}, the diameter is $\leq 2n+1$; if every element is $s\,y$ with $s \in S$ and y a product of at most n factors $s^{-1}t$, the diameter is $\leq 2n+2$.

The records. The covering conditions are finite checks; for both graphs the Lean files contain, for every group element, a quadruple from S that represents it (`lean/BipartiteDD/Haar/Cert1152.lean`, `Cert2662.lean`), checked by the kernel.

- $(7,5)$, 1152 vertices: $G = S_4 \times \mathbb{Z}_{24}$ (order 576), product $(f,x)(h,y) = (f \circ h, x+y)$, and

  $$S = \{(\mathrm{id},0),\ ((0\,1)(2\,3),23),\ ((0\,3\,2),0),\ ((0\,2),18),\ ((0\,1\,2\,3),18),\ ((1\,2\,3),17),\ ((1\,2),3)\},$$

  permutations of $\{0,1,2,3\}$ in cycle notation (Lean lists them in one-line form $[f(0),f(1),f(2),f(3)]$).
- $(9,5)$, 2662 vertices: G the Heisenberg group mod 11, $(a,b,c)(a',b',c') = (a+a', b+b', c+c'+ab')$ (the matrix $\left(\begin{smallmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{smallmatrix}\right)$ over $\mathbb{Z}_{11}$), and

  $$S = \{(0,0,0),(8,6,10),(7,9,3),(10,1,8),(4,4,5),(5,5,8),(6,5,4),(1,0,9),(6,10,8)\}.$$
- $(8,5)$, 1764 vertices: $G = \mathrm{AGL}(1,7) \times \mathbb{Z}_{21}$ as $(x,y,z) \in \mathbb{Z}_7 \times \mathbb{Z}_6 \times \mathbb{Z}_{21}$, $(x,y,z)(x',y',z') = (x+3^y x', y+y', z+z')$, $S = \{(0,0,0),(0,4,5),(6,3,18),(6,5,6),(2,4,1),(2,2,16),(4,5,12),(2,3,15)\}$. (This cell's current record, 2048, comes from a different lift construction not covered here.)

Cells. $(7,5)$: 1152 (2013: 1134). $(9,5)$: 2662 (2013: 2496). $(8,5)$: 1764 (2013: 1710; superseded). All regular.

Status. Paper proof (above). Computer-checked: the record files pass the lab's checkers, and for this note all three graphs were rebuilt from (G,S) exactly as printed above (order 1152, 2662, 1764; diameter 5, from the eccentricities of (e,A) and (e,B), which suffice because right multiplication is transitive on each side). Lean-checked: the lemma for every group (`Haar.haar_ediam_le_five`, `haar_isBipartiteDegDiam`) and the record graphs (`nb_7_5_v2` for 1152, `nb_9_5_v6` for 2662, `nb_8_5_v2` for 1764).

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9. Exactly what is Lean-checked

All theorems below are in `lean/BipartiteDD` (Lean 4.34.1, Mathlib commit `d13f23b7`). For each, `#print axioms` (files `lean/Axioms.out.txt`, `lean/RegularAxioms.out.txt`, `lean/BipartiteDD/*/Axioms.out.txt`) lists only `propext`, `Classical.choice`, `Quot.sound` (a few helper lemmas list a subset). `grep` finds no `sorry`, `admit`, `axiom` or `native_decide` in any `.lean` file, and no `implemented_by`, `extern`, `unsafe` or `ofReduceBool`; finite checks use `decide` / `decide +kernel`, which the kernel checks.

The graph predicate. `IsBipartiteDegDiam G d D` means: G is 2-colourable, every vertex has at most d neighbours, and the diameter (`ediam`, which is ∞ for a disconnected graph) is at most D. `IsBipartiteRegDiam G Δ D` is the same with every vertex of degree exactly Δ. Every record statement has the form "there is a graph on `Fin N` with this property", so the vertex count is exactly N; diameters are upper bounds.

Two conventions to know. 1. `GQst P L s t` (the generalized-quadrangle hypothesis) records: lines have $s+1$ points, points are on $t+1$ lines, two points are on at most one line, and for an anti-flag (p, L) *some* point of L is collinear with p (the existence half of the GQ axiom). Theorems that *assume* `GQst` are therefore slightly stronger than their paper statements; theorems that *conclude* `GQst` (`ASq.isGQst`, `SingerC.W3.isGQst`) prove less than "is a GQ", but exactly what Theorem B needs. 2. AS(q), $W(3,q)$ and $H(q)$ are each given by the coordinate model used in this note; Lean does not prove that the model is isomorphic to the classical object (we cite [P71], [PT], [T59], [VM98] for that). The generalized polygon predicate `GenPolygonGraph G n` is: 2-colourable, girth $2n$, diameter n, every vertex has two distinct neighbours.

Section 1 (doubling, hat lemma). `Doubling.delorme_levelled`, `isBipartiteDegDiam_levelled`, `card_levelled_levels`: Theorem 1.2(a)–(c) for any finite graph that is bipartite of maximum degree $\le \Delta$ and diameter $\le D$, with (L0), (L1), (C_D) as hypotheses, giving maximum degree $\le \Delta + m - 1$, diameter $\le D$ and the exact count. `Doubling.condC_of_forestDiam` (any graph with all degrees ≥ 2) and `GenPolygonGraph.condC_iff_forestDiam` (both directions of the hat lemma); `delorme_levelled_polygon` (Corollary 1.4, "if"). Theorem 1.2(d) is not formalised.

Section 2 (fourth hat). `Plus2.plus2_regular`: for every finite projective plane (Mathlib's `ProjectivePlane`, not necessarily Desarguesian) with a frame (a line ℓ, a point $A \notin \ell$, five distinct points of ℓ), there is a $(q+2)$-regular bipartite graph on $2(q^2+q+6)$ vertices of diameter ≤ 3. `plus2_regular_of_order`: the same for every finite projective plane of order $q \ge 4$. (`plus2`, `plus2_of_order`: the non-regular version.) Records: `nb_11_3` (192), `nb_13_3` (276), `nb_15_3` (376), explicit graphs. A variant that keeps every incidence of the plane, for $q \ge 3$, is `Plus2Zero.zero_isBipartiteDegDiam` (not described in this note).

Section 3 (Theorem B). For every finite `GQst P L s t` and multiplicities $w : P \to \mathbb{N}$, $m : L \to \mathbb{N}$ with line sums of w at most a and point sums of m at most b: `GQST.gqst` (all stand-ins original) gives a bipartite graph on $|P| + |L| + \sum w + \sum m$ vertices with maximum degree $\le \max(s+1+a, t+1+b)$ and diameter ≤ 4; `GQST.standin_gqst` is the same with arbitrary stand-ins and "each stand-in serves at most b line copies"; `GQST.regular_copy_graph`, `regular_copy_graph_fin`: with exact sums ($\Delta - s - 1$ and $\Delta - t - 1$) the graph can be made Δ-regular with diameter ≤ 4 (Corollary 3.3). Record: `nb_7_4` (the 359 file `records/bipartite_dd/d7D4/359-spec-d7d4-g017-c11.json`).

Section 4 (Theorem A). For every finite field $\mathbb{F}$ of odd characteristic, with $q = |\mathbb{F}|$: `ASq.isGQst` (the coordinate model of §4.1 is `GQst` of order $(q-1, q+1)$), `ASq.weighted_two_ovoid` (Theorem 4.1's line count, for ε a non-square and $c_1 c_2 = 1/(4\varepsilon)$), `ASq.bound` and `ASq.bound_regular` (a $(q+2)$-regular bipartite graph on $2q^2(q+2)$ vertices of diameter ≤ 4). Records: `nb_11_4_v2` (1782), `nb_13_4_v2` (3146), `nb_15_4_v2` (5070).

Section 5 (Proposition C, Corollary C1). For every field E of order q^4, $q = p^e$ odd, with a chosen element g of order $q^4 - 1$ (a `SingerC.Setting`): `SingerC.Setting.propC_subspace` and `SingerC.propC` (every totally isotropic line of $W(3,q)$, in the model of §5.1, has exactly 2 points in $C_a \cup C_{q-a}$, for every $a \le q$), `SingerC.W3.isGQst`, `SingerC.W3.spread_exists_unique` (the lines yK form a spread), `SingerC.corC1` (for every even $j \ge 2$, a bipartite graph on $2(q^2+1)(q+1+j)$ vertices with maximum degree $\le q+1+j$ and diameter ≤ 4, built in $W(3,q)$ itself) and `SingerC.Setting.regular_W3` (the regular graph of Corollary 5.2). Such fields and elements exist for every odd prime power, but the library constructs one only for $q = 13$ (`setting13`), giving `SingerC.regular5440` (a 16-regular bipartite graph on 5440 vertices of diameter ≤ 4). Record: `nb_16_4_v2` (the 5440 file).

Section 6 (Theorem D). For every finite field of odd characteristic: `Hexagon.genPolygonGraph_hexagon`

($H(q)$ in the coordinates of §6.1 is a generalized hexagon in the graph sense); `HexagonD.theorem4` (there are c, c_0, c_1 with $-c$ a non-square and G zero-free such that W of Theorem 6.5 is a whisker set with $|W|+q^2 = q^4$ and every free point on exactly one line of W); `HexagonD.whiskers_bound_hexagon` (Theorem 6.3 for any whisker set); `HexagonD.theoremD`, `theoremD_two` (Corollary 6.8 for every $m \geq 1$, resp. $m = 2$, with the exact vertex counts); `HexagonD.exists_goodCubic` (Lemma 6.4); `HexagonD.Identity.key_identity` (Lemma 6.7, over any field) and `nu_eq_of_mem_sline` (Lemma 6.6(d)). Even q: `HexagonEven.genPolygonGraph_hexagon_even` (any finite field), `theorem4_even`, `theoremD_even` ($q \geq 4$); `theoremD_q2` and `q2_exact` ($q = 2$, $m = 2$, the 12 explicit whiskers). Record: `nb_7_6` (the 9592 file).

Section 7 (Theorem R). `HexagonD.theoremR`, `theoremR_two` (odd characteristic, $m \geq 2$, resp. $m = 2$): a $(q + m)$-regular bipartite graph of diameter ≤ 6 on $2(q + 1)(q^4 + q^2 + 1 + (m - 1)(q^3 + m + 1))$ vertices. `HexagonEven.theoremR`, `theoremR_two` (characteristic 2, $q \geq 4$), `theoremR_q2` ($q = 2$, $m = 2$, 192 vertices).

Section 8 (Haar). For every group G and finite nonempty S: `Haar.haar_ediam_le_five`, `haar_isBipartiteDegDiam` (Lemma 8.2), `haar_ediam_le_odd`, `haar_ediam_le_even`. Records `nb_7_5_v2` (1152), `nb_9_5_v6` (2662), `nb_8_5_v2` (1764).

Not formalised. Theorem 1.2(d); the identification of the coordinate models with the classical $\mathrm{AS}(q)$, $W(3, q)$, $H(q)$; the converse of the Haar lemma; that the 359 graph decomposes as in §3 (Lean checks the graph, not its description); a general existence lemma for `SingerC.Setting`; Theorem R's completion is proved in Lean but, as on paper, non-constructively.

Build provenance. A full `lake build` of the library passed on 2026-10-20 (`lean/CHECK.txt`, 9115 jobs; heavy record modules rebuilt one at a time). Several modules were added after that date (`HexagonD/Regular`, `HexagonEven`, `Haar/GraphD`, on which the current `Haar.Main` depends). **For this note (2026-10-25) the 23 modules behind every general theorem cited above were rebuilt from a fresh clone** (Lean 4.34.1, Mathlib `d13f23b7`, one `lake build <module>` at a time): all exit 0, including `HexagonD.Regular.Final`, `HexagonEven.ThmD`, `HexagonEven.Q2`, `HexagonEven.Reg.Final`, `Haar.Main` and the three Haar records, `GQST.StandinMain` and `SingerC.Regular5440`. A `#print axioms` run on the 48 theorem names cited in this section gives `[propext, Classical.choice, Quot.sound]` for every one (`research/theorems/polish/lean/PolishAxioms.lean`, `.out`, and `BUILD.txt` with the per-module log), and the grep for `sorry/admit/axiom/native_decide` in `lean/BipartiteDD/**.lean` finds nothing. The explicit record graphs were rebuilt the same day: `Records.lean`, `Records9592.lean` and all 91 modules they import (one at a time; four first attempts were killed for lack of memory and passed on retry), and `#print axioms` on `nb_11_3`, `nb_13_3`, `nb_15_3`, `nb_7_4`, `nb_11_4_v2`, `nb_13_4_v2`, `nb_15_4_v2`, `nb_16_4_v2`, `nb_7_6` gives the same three axioms (`polish/lean/RecordAxioms.lean`, `.out`).

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Appendix A. The $(7, 4)$ example in coordinates

$Q(4, 5)$: the quadric $X_0^2 + X_1X_2 + X_3X_4 = 0$ of PG$(4, 5)$, with its 156 points and 156 lines (points are written with first non-zero coordinate 1). Ovoid $\mathcal{O} = Q(4, 5) \cap \{X_4 = 2X_3\}$ (-2 is a non-square mod 5), an elliptic quadric with 26 points. The structure:

- the incidence graph of $Q(4, 5)$ (312 vertices, all of degree 6);
- one clone o' of each $o \in \mathcal{O}$, joined to the 6 lines on o (26 vertices; every line gains exactly one clone, as $\mathcal{O}$ is an ovoid);
- 21 shadow lines, one for each line below, joined to its six points, except that a starred point is replaced by its clone (the stand-in).

```
 1  (0:0:0:0:1) (0:1:0:0:0)  (0:1:0:0:1) (0:1:0:0:2)  (0:1:0:0:3)  (0:1:0:0:4)
 2  (0:0:0:1:0) (0:1:0:0:0)* (0:1:0:1:0) (0:1:0:2:0)  (0:1:0:3:0)  (0:1:0:4:0)
 3  (0:0:1:0:0) (1:0:0:2:2)  (1:0:1:2:2) (1:0:2:2:2)  (1:0:3:2:2)  (1:0:4:2:2)
 4  (0:0:1:0:1) (1:2:0:3:3)  (1:2:1:3:4) (1:2:2:3:0)  (1:2:3:3:1)  (1:2:4:3:2)
 5  (0:0:1:0:2) (1:1:0:2:2)  (1:1:1:2:4) (1:1:2:2:1)  (1:1:3:2:3)  (1:1:4:2:0)
 6  (0:0:1:0:4) (1:3:0:3:3)  (1:3:1:3:2) (1:3:2:3:1)  (1:3:3:3:0)  (1:3:4:3:4)
 7  (0:0:1:1:0) (1:4:0:4:1)  (1:4:1:0:1) (1:4:2:1:1)  (1:4:3:2:1)  (1:4:4:3:1)
 8  (0:0:1:2:0) (1:4:0:3:3)  (1:4:1:0:3) (1:4:2:2:3)  (1:4:3:4:3)  (1:4:4:1:3)
 9  (0:0:1:3:0) (1:2:0:4:1)  (1:2:1:2:1) (1:2:2:0:1)  (1:2:3:3:1)* (1:2:4:1:1)
10  (0:0:1:4:0) (1:4:0:1:4)  (1:4:1:0:4) (1:4:2:4:4)  (1:4:3:3:4)  (1:4:4:2:4)
11  (0:1:1:1:4) (1:0:3:4:1)  (1:1:4:0:0) (1:2:0:1:4)  (1:3:1:2:3)  (1:4:2:3:2)
12  (0:1:1:2:2) (1:0:0:1:4)  (1:1:1:3:1) (1:2:2:0:3)  (1:3:3:2:0)  (1:4:4:4:2)
13  (0:1:1:3:3) (1:0:2:3:3)  (1:1:3:1:1) (1:2:4:4:4)  (1:3:0:2:2)  (1:4:1:0:0)
14  (0:1:1:4:1) (1:0:2:4:1)  (1:1:3:3:2) (1:2:4:2:3)  (1:3:0:1:4)  (1:4:1:0:0)*
15  (0:1:2:1:3) (1:0:3:3:3)  (1:1:0:4:1) (1:2:2:0:4)  (1:3:4:1:2)  (1:4:1:2:0)
16  (0:1:2:2:4) (1:0:3:1:4)  (1:1:0:3:3) (1:2:2:0:2)  (1:3:4:2:1)  (1:4:1:4:0)
17  (0:1:3:2:1) (1:0:1:3:3)  (1:1:4:0:4) (1:2:2:2:0)  (1:3:0:4:1)  (1:4:3:1:2)
18  (0:1:3:3:4) (1:0:4:1:4)  (1:1:2:4:3) (1:2:0:2:2)  (1:3:3:0:1)  (1:4:1:3:0)
19  (0:1:4:1:1) (1:0:4:3:3)  (1:1:3:4:4) (1:2:2:0:0)  (1:3:1:1:1)  (1:4:0:2:2)
20  (0:1:4:2:3) (1:0:0:3:3)  (1:1:4:0:1) (1:2:3:2:4)  (1:3:2:4:2)  (1:4:1:1:0)
21  (0:1:4:4:4) (1:0:0:4:1)  (1:1:4:3:0) (1:2:3:2:4)* (1:3:2:1:3)  (1:4:1:0:2)
```

What to check by hand (Theorem 3.2(b) with $s = t = 5$, $\Delta = 7$): each listed line lies on the quadric; the 21 lines are pairwise disjoint except at the four ovoid points $(0:1:0:0:0)$, $(1:2:3:3:1)$, $(1:4:1:0:0)$, $(1:2:3:2:4)$, each on exactly two of them (lines 1 and 2, 4 and 9, 13 and 14, 20 and 21), and on the second of the two the point is replaced by its clone. So every old point and every clone serves at most one shadow line. Degrees: old lines $6+1=7$; old points 6 or 7; clones 6 or 7; shadow lines 6. Order $156+156+26+21=359$; diameter 4 by Theorem 3.2(a). The list was produced by `research/theorems/polish/q4_359.py` (HiGHS proves 21 optimal for this family), which also rebuilds the graph and checks maximum degree 7 and diameter 4. Lean checks the record file of the same order (`nb_7_4`), which has the same shape (Section 3), not this particular list.